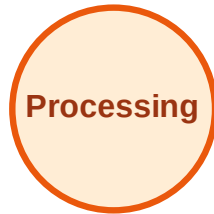


BFS Traversal

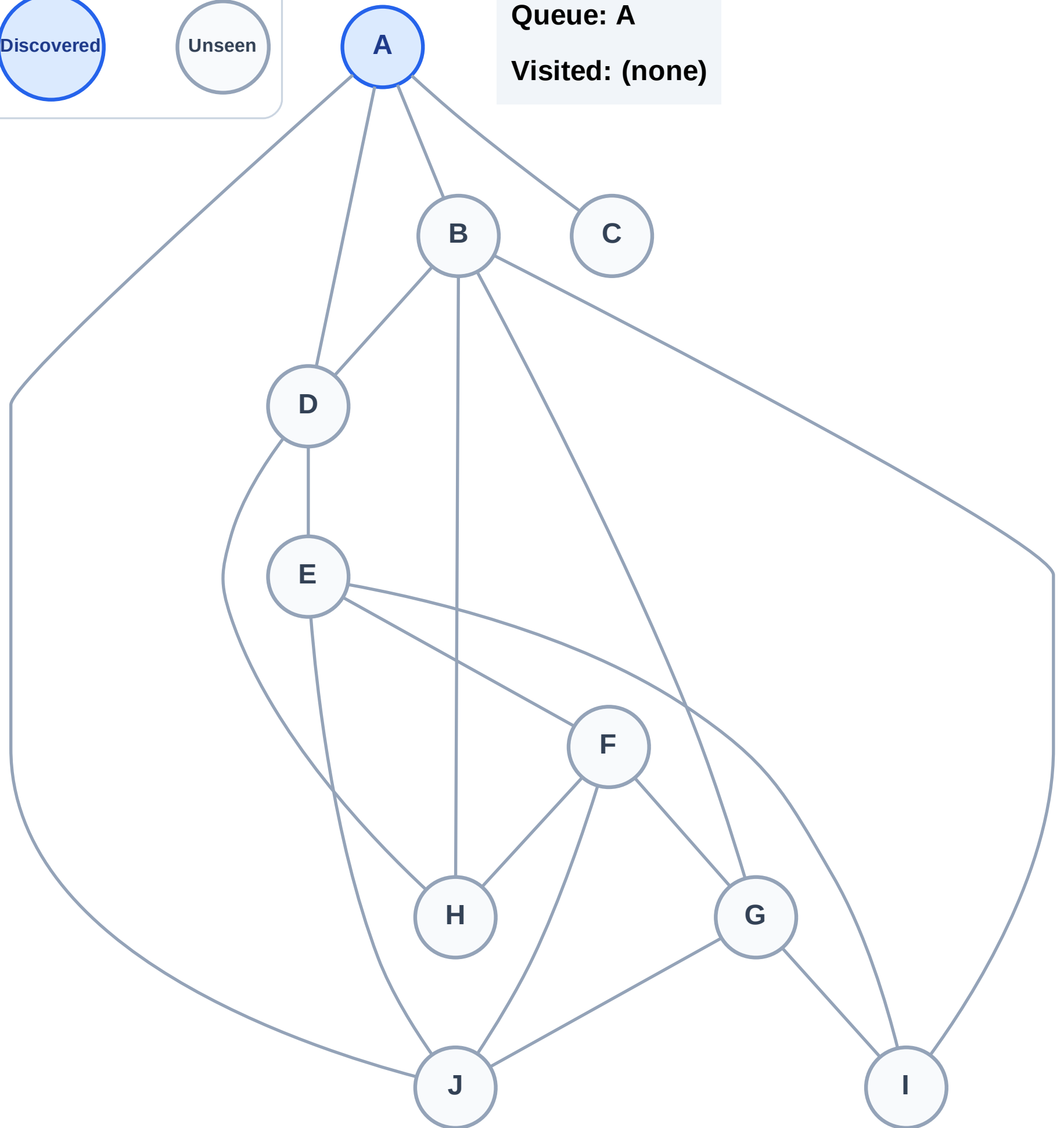
Step 1: Start at A

Legend



Queue: A

Visited: (none)



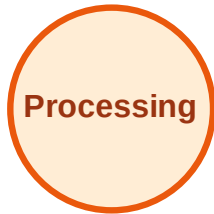
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



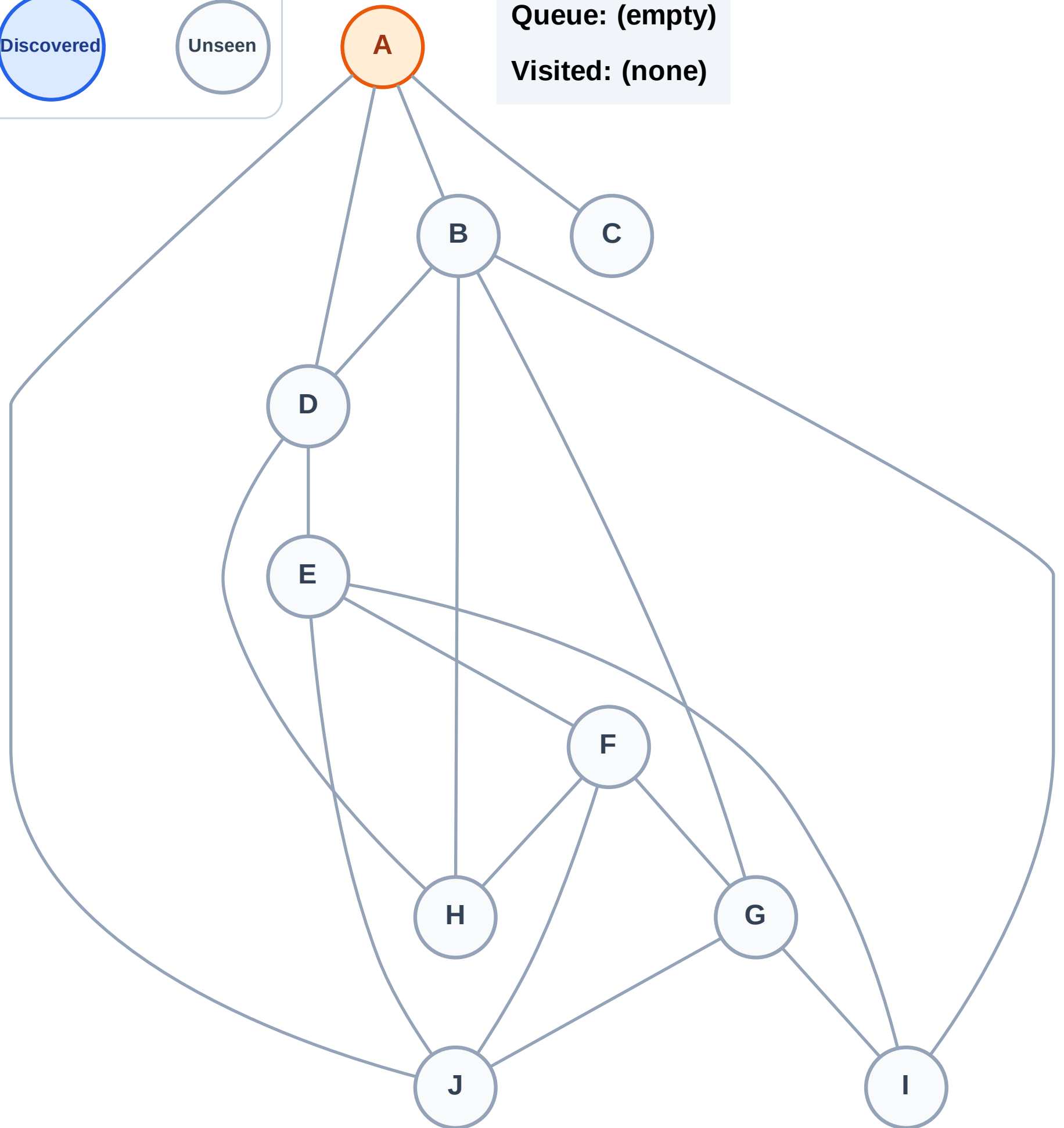
Discovered



Unseen

Queue: (empty)

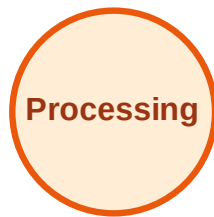
Visited: (none)



BFS Traversal

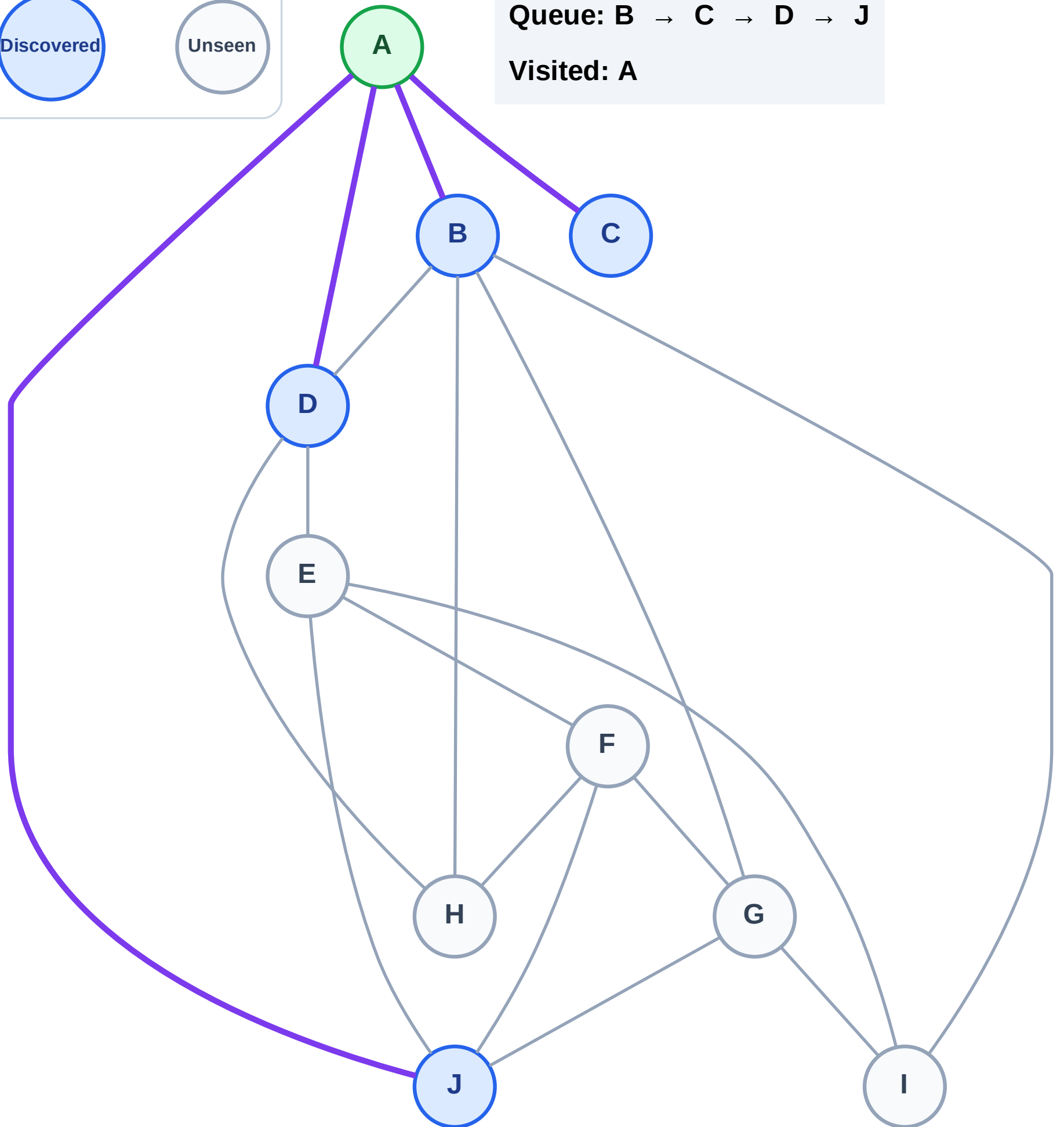
Step 3: Discover B, C, D, J from A

Legend



Queue: B → C → D → J

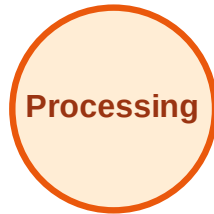
Visited: A



BFS Traversal

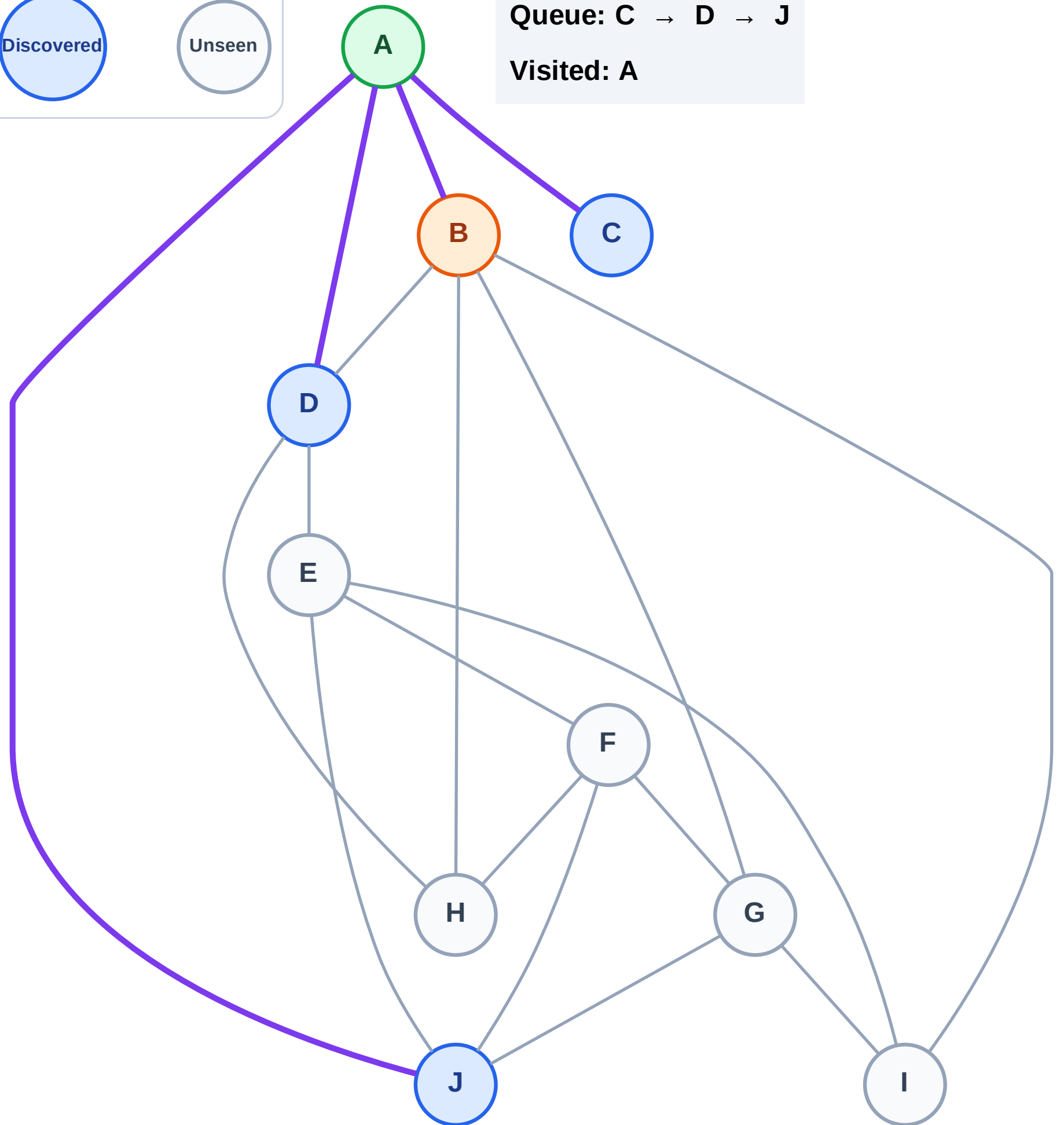
Step 4: Process node B

Legend



Queue: C → D → J

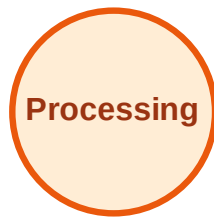
Visited: A



BFS Traversal

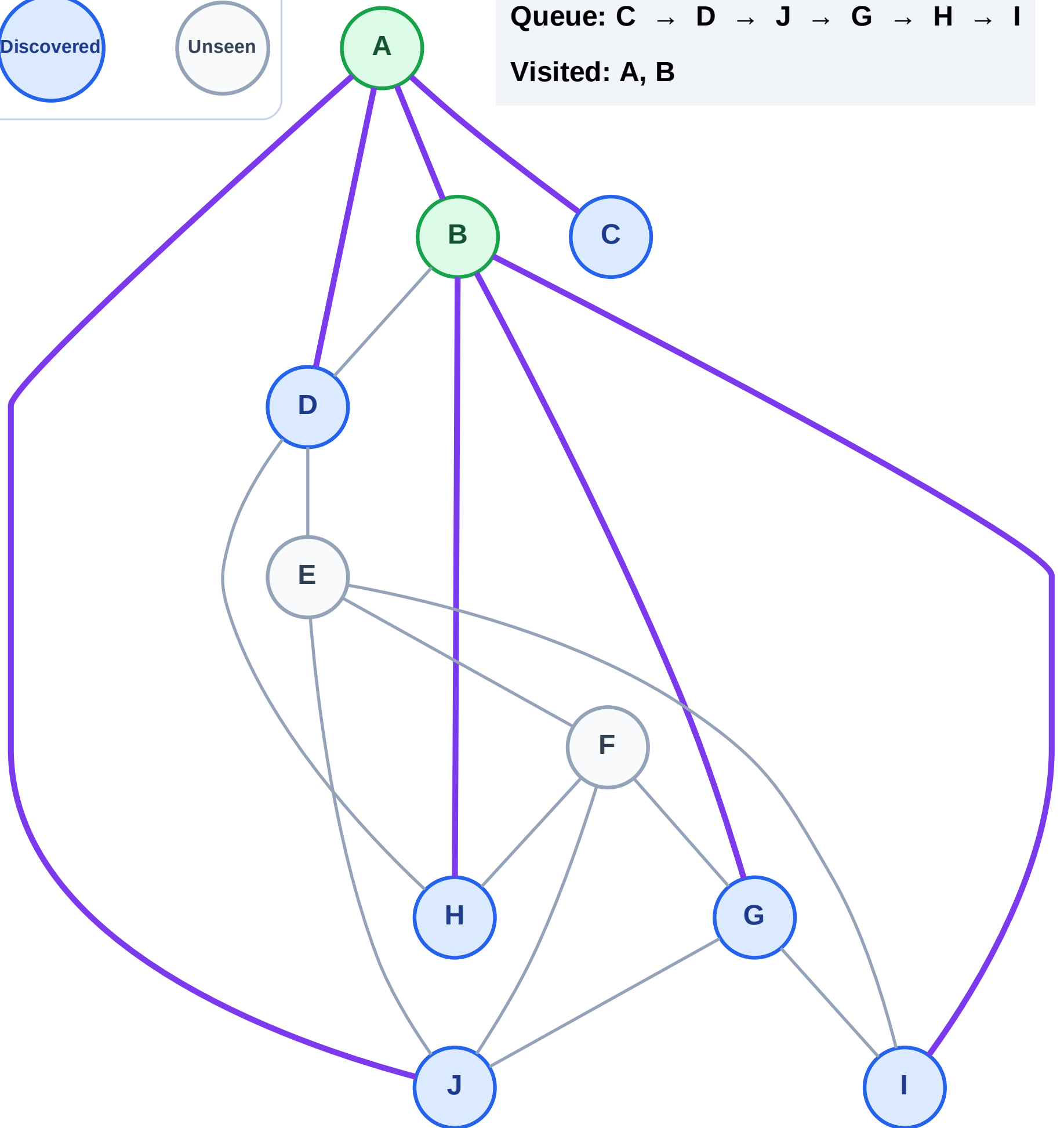
Step 5: Discover G, H, I from B

Legend



Queue: C → D → J → G → H → I

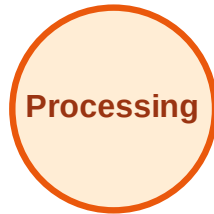
Visited: A, B



BFS Traversal

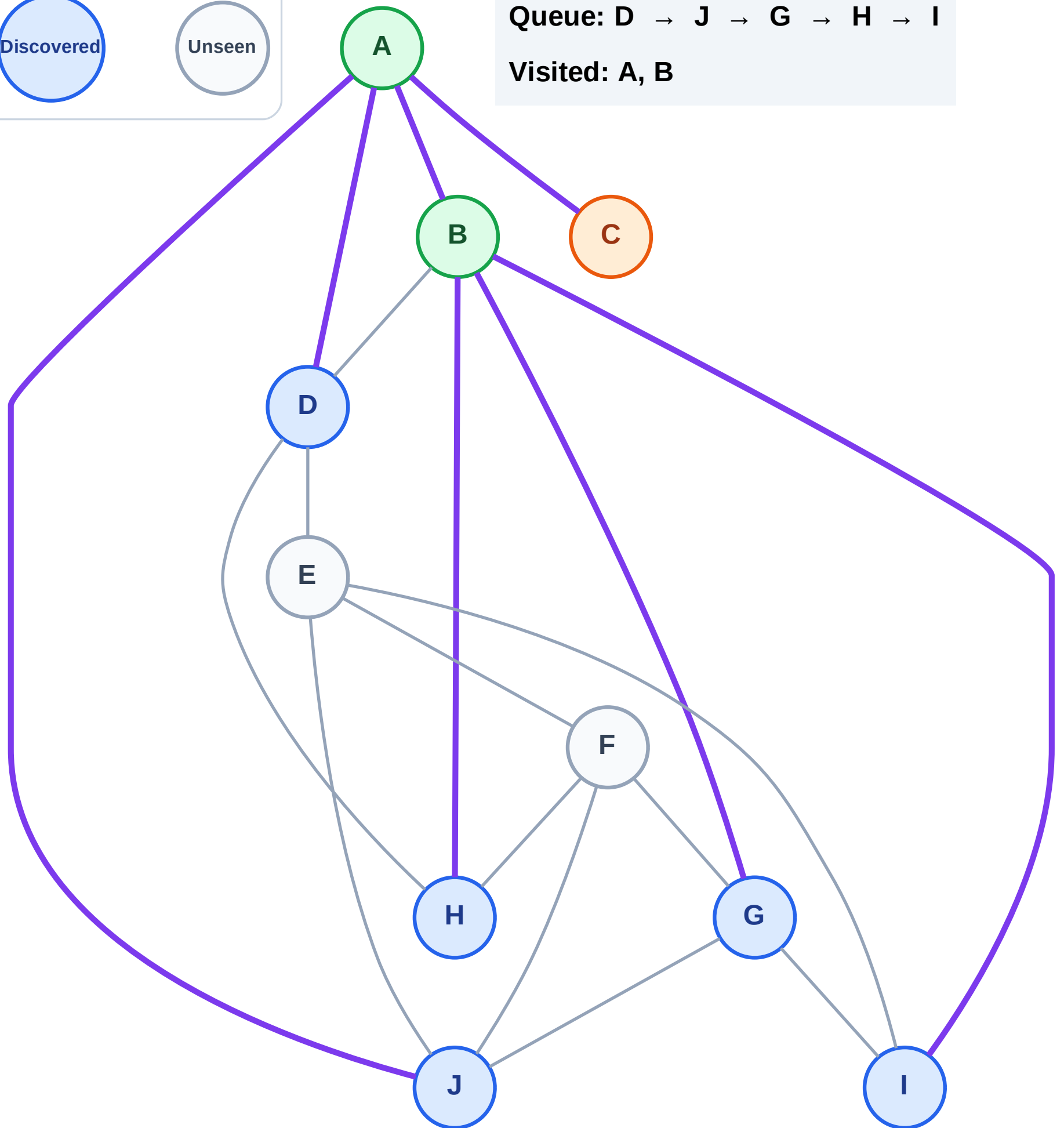
Step 6: Process node C

Legend



Queue: D → J → G → H → I

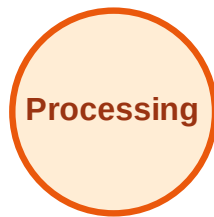
Visited: A, B



BFS Traversal

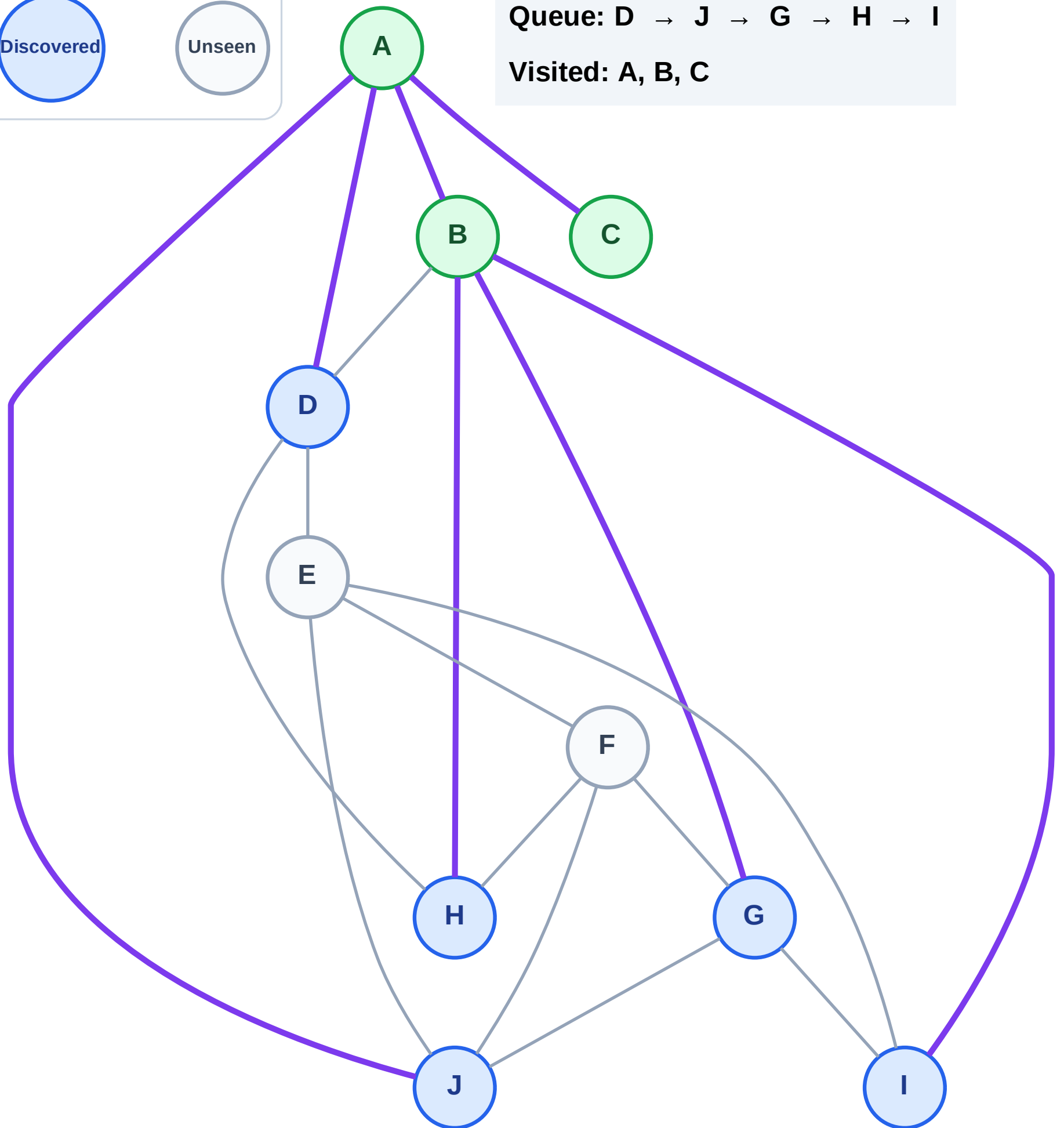
Step 7: No new neighbors from C

Legend



Queue: D → J → G → H → I

Visited: A, B, C



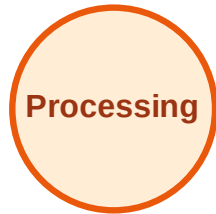
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



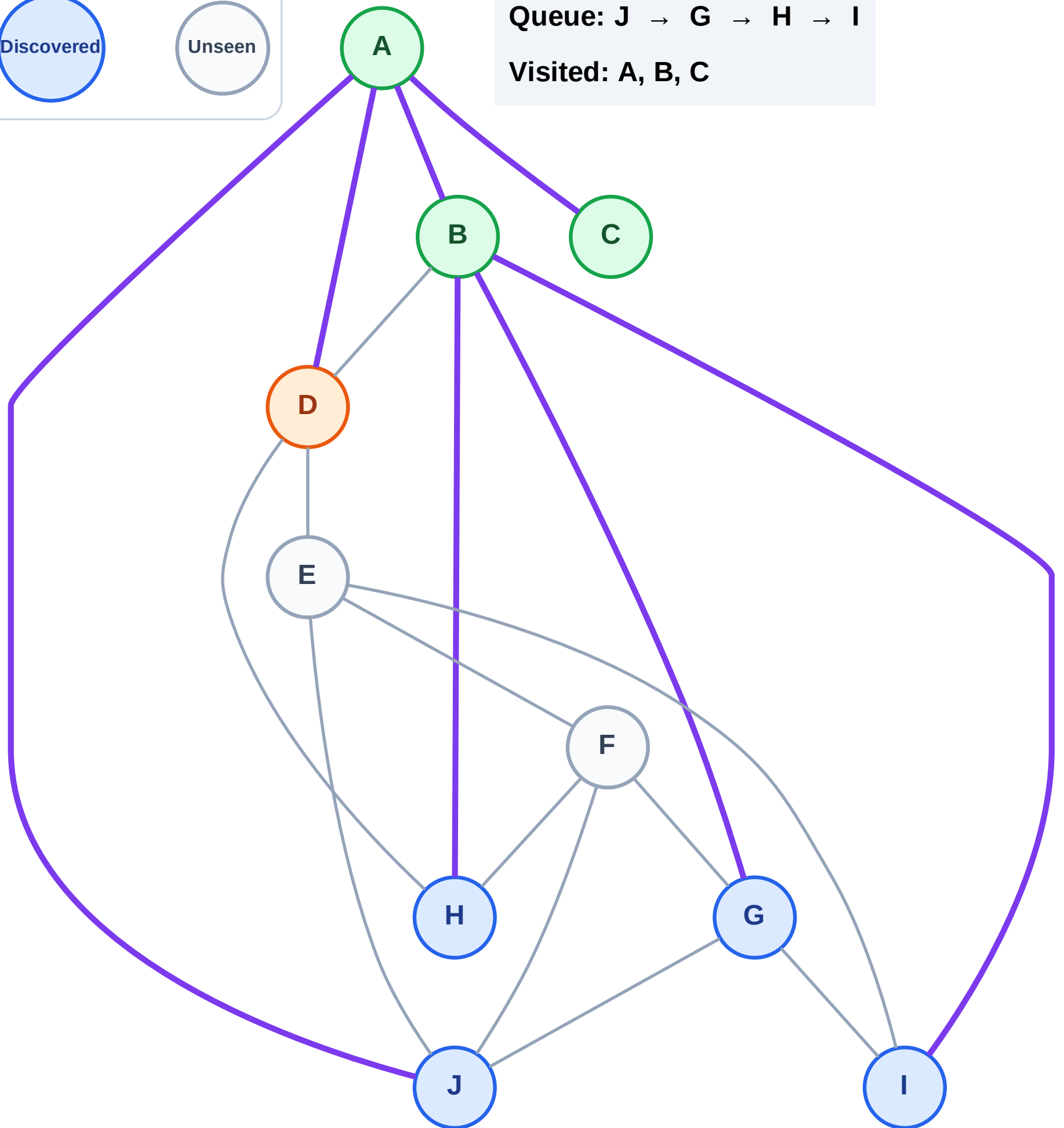
Discovered



Unseen

Queue: J → G → H → I

Visited: A, B, C



BFS Traversal

Step 9: Discover E from D

Legend

Complete

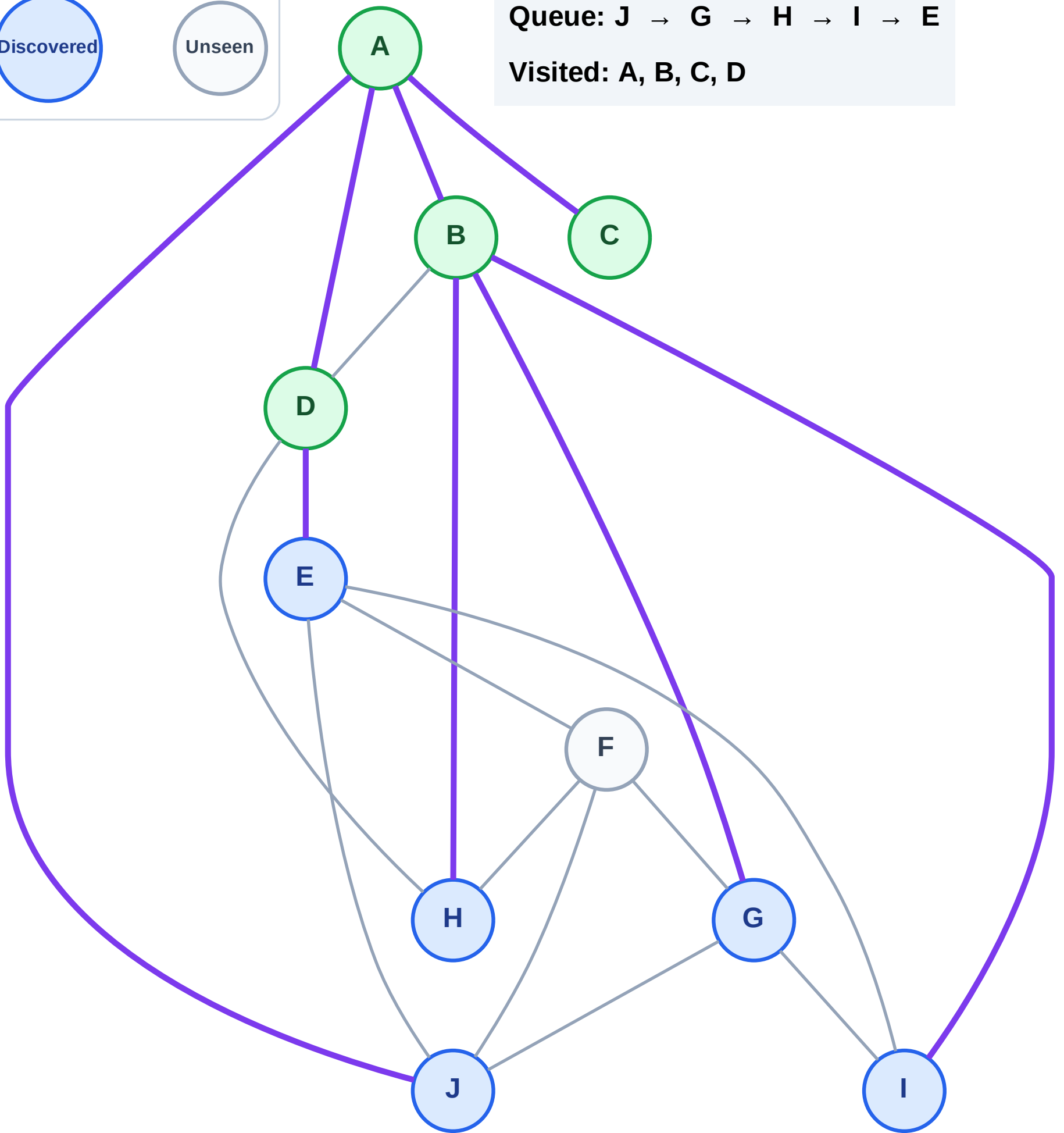
Processing

Discovered

Unseen

Queue: J → G → H → I → E

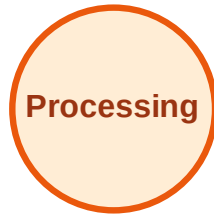
Visited: A, B, C, D



BFS Traversal

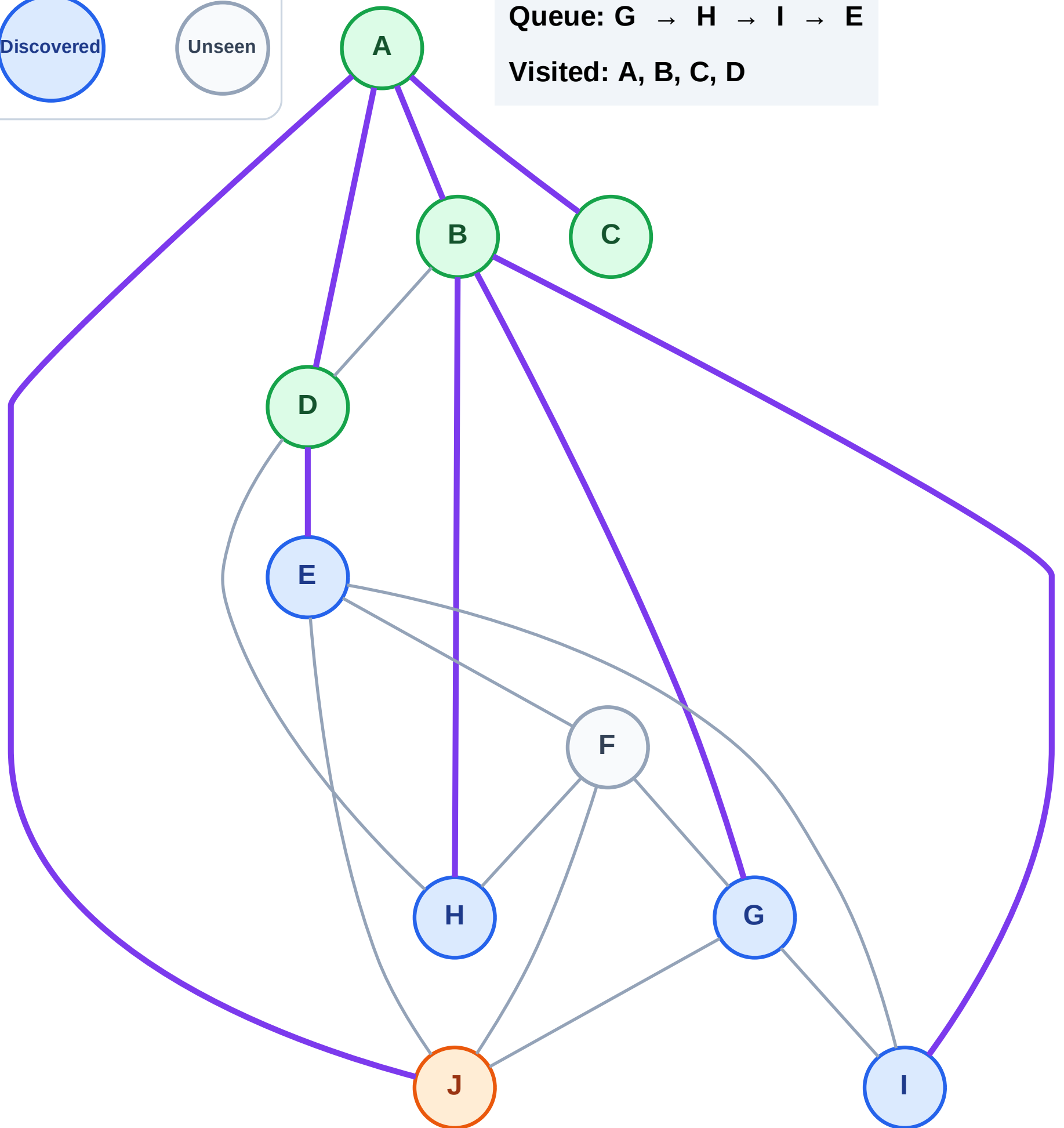
Step 10: Process node J

Legend



Queue: G → H → I → E

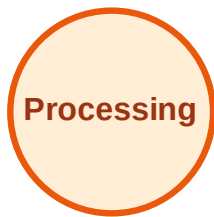
Visited: A, B, C, D



BFS Traversal

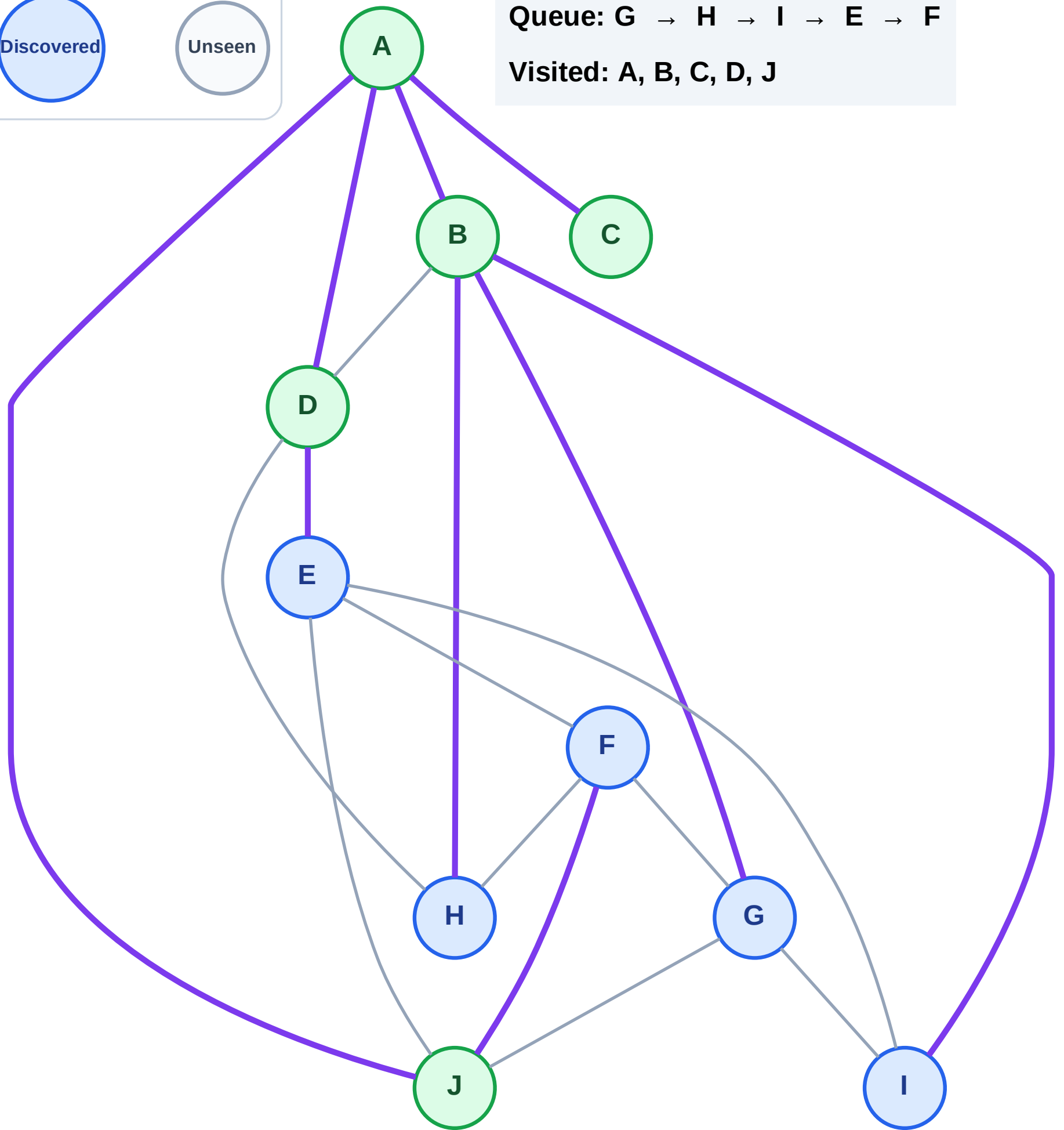
Step 11: Discover F from J

Legend



Queue: G → H → I → E → F

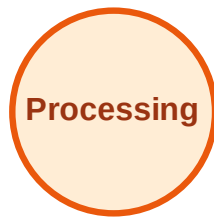
Visited: A, B, C, D, J



BFS Traversal

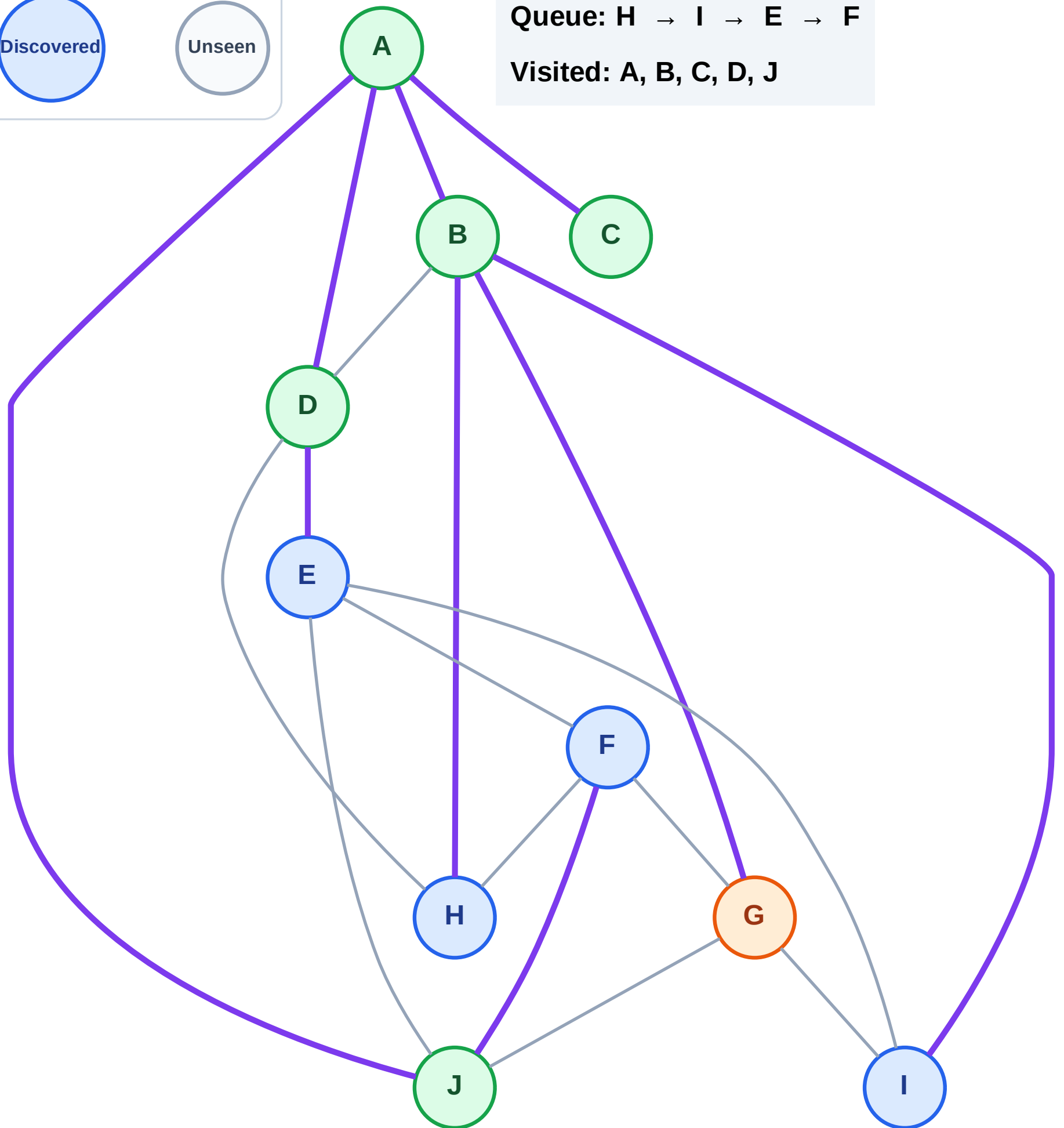
Step 12: Process node G

Legend



Queue: H → I → E → F

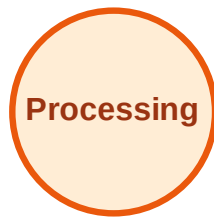
Visited: A, B, C, D, J



BFS Traversal

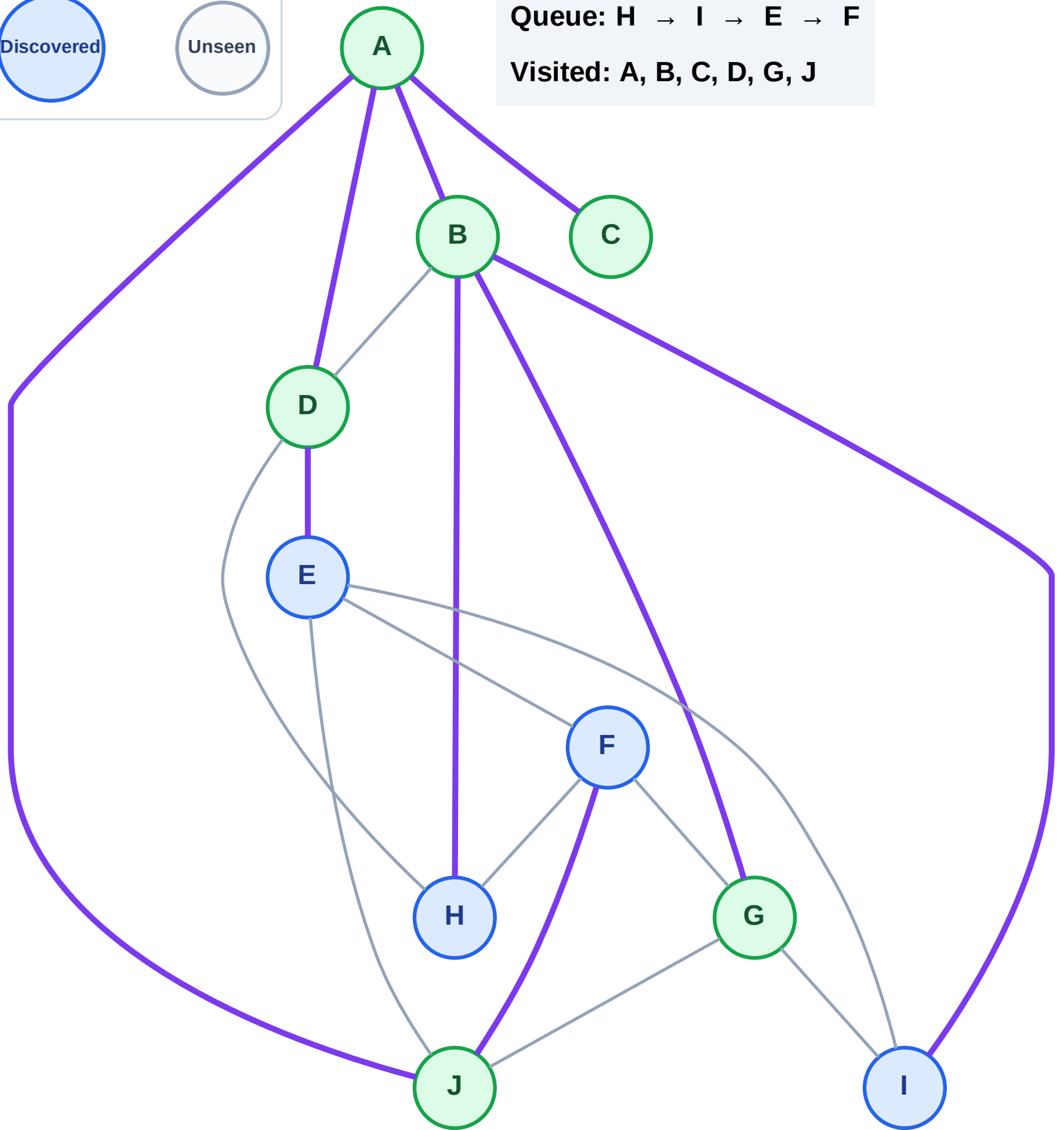
Step 13: No new neighbors from G

Legend

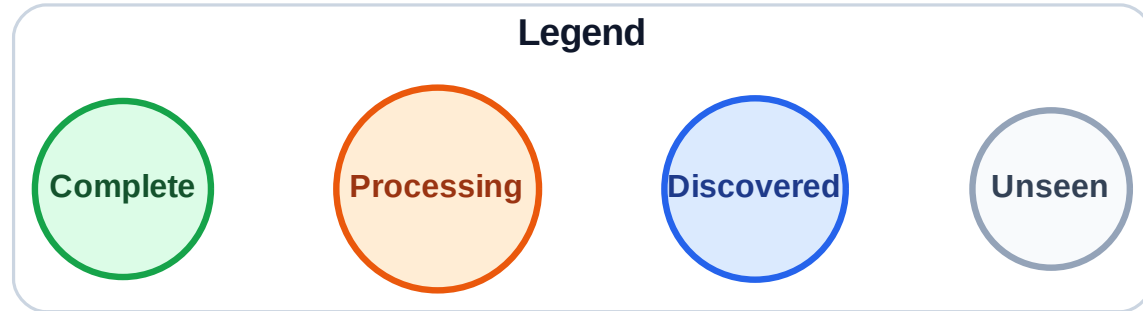


Queue: H → I → E → F

Visited: A, B, C, D, G, J

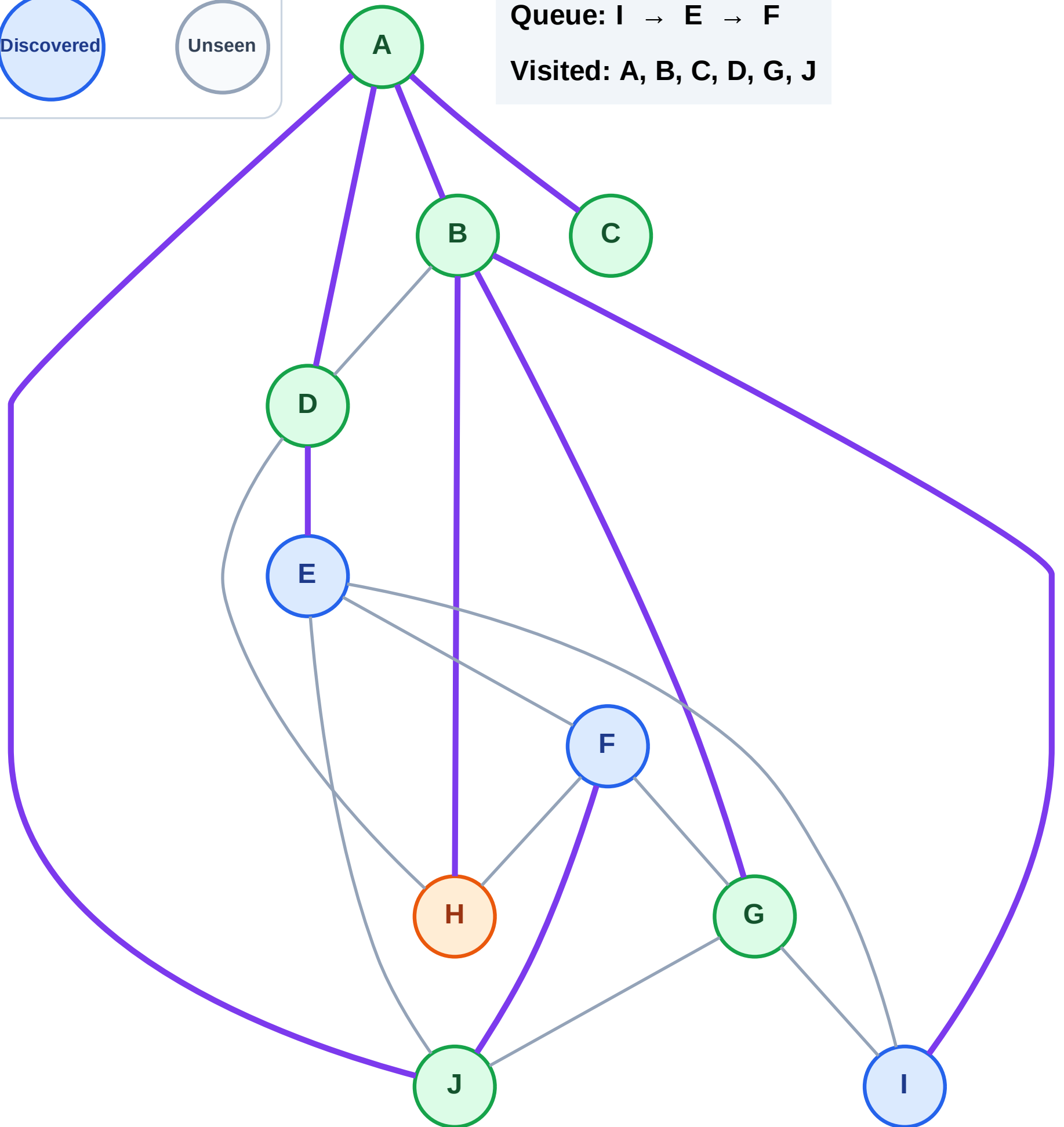


Step 14: Process node H



Queue: I → E → F

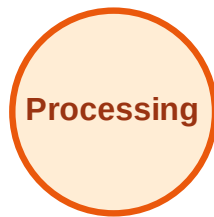
Visited: A, B, C, D, G, J



BFS Traversal

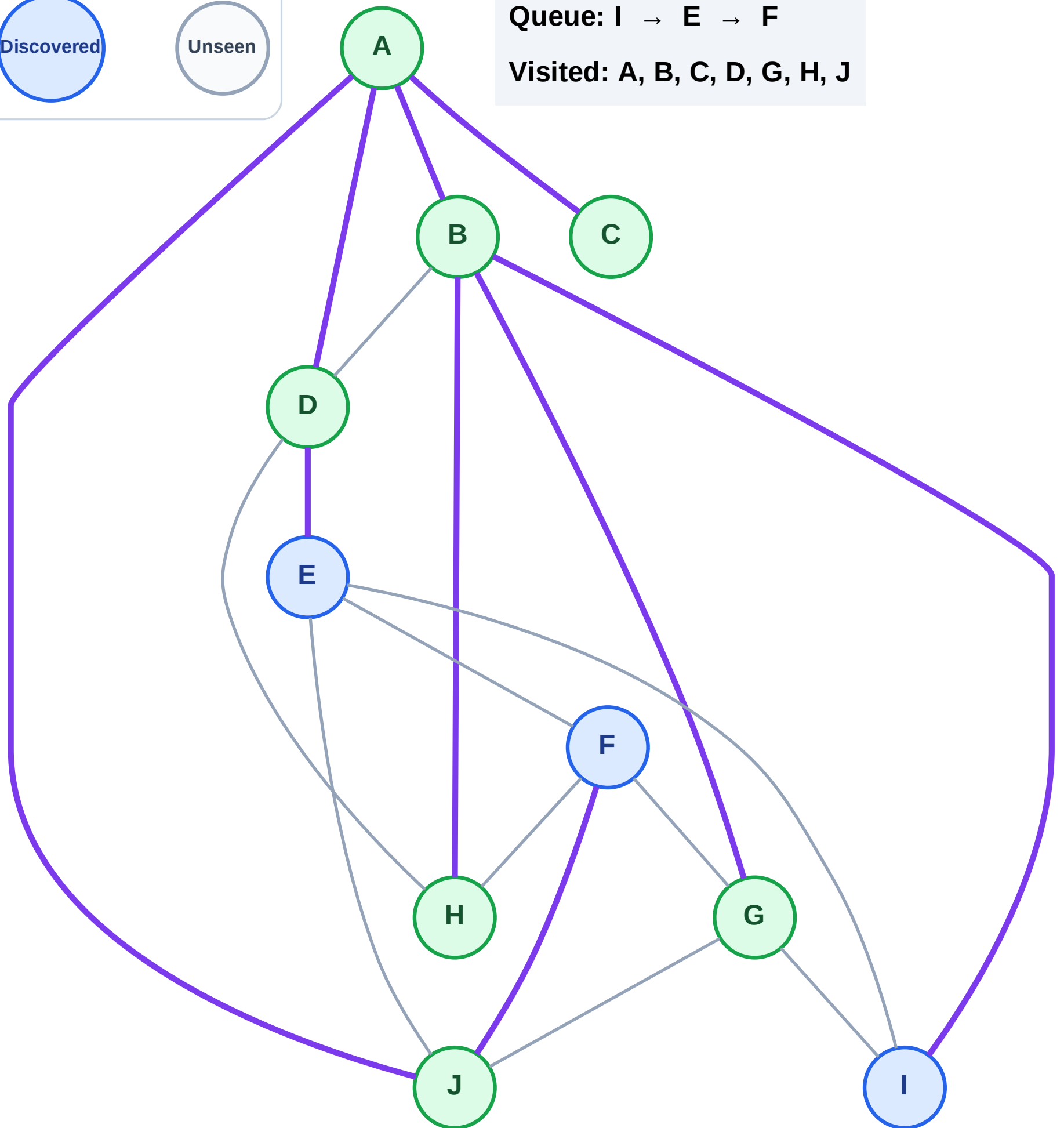
Step 15: No new neighbors from H

Legend



Queue: I → E → F

Visited: A, B, C, D, G, H, J



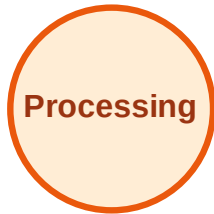
BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



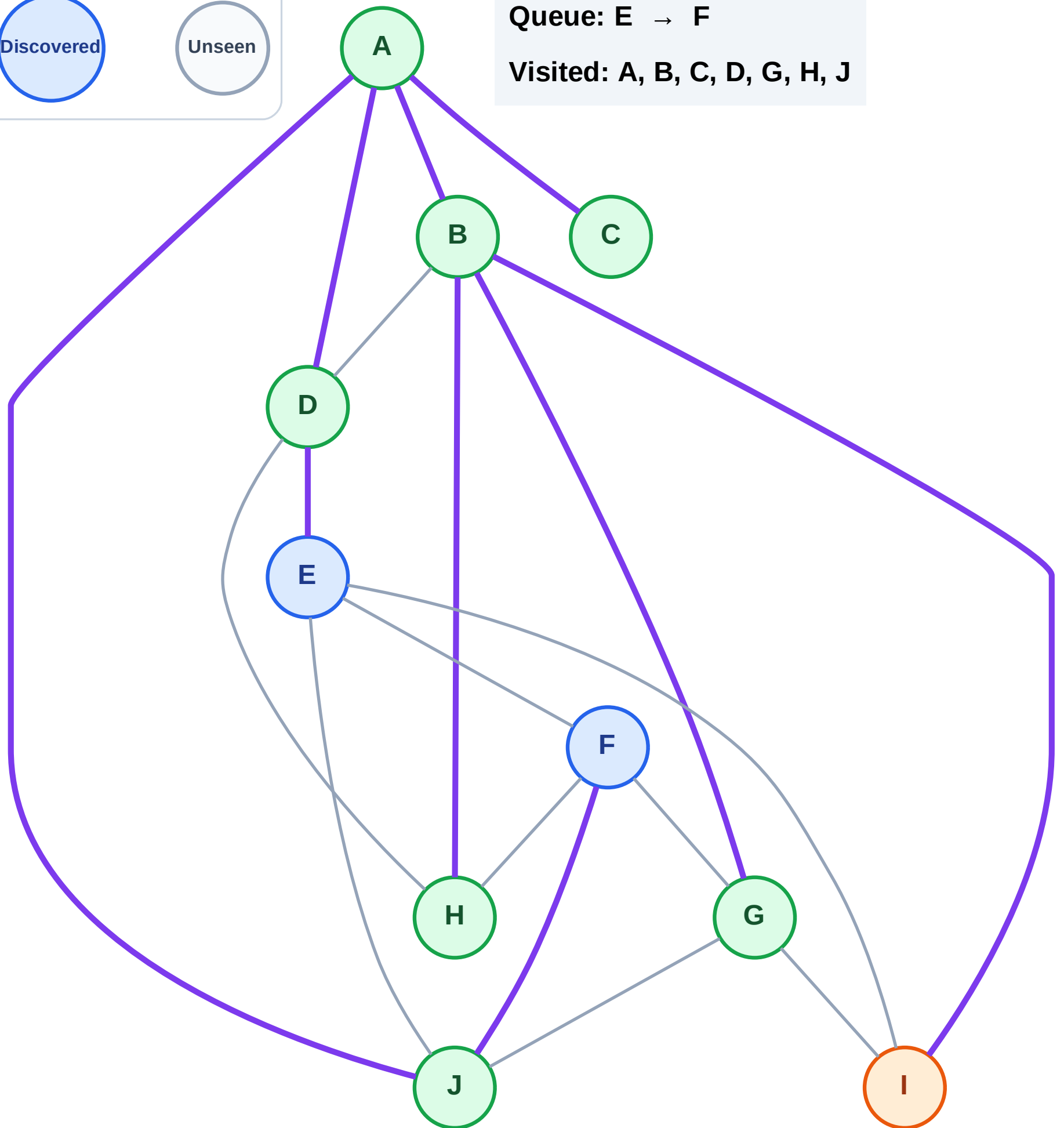
Discovered



Unseen

Queue: E → F

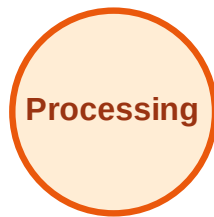
Visited: A, B, C, D, G, H, J



BFS Traversal

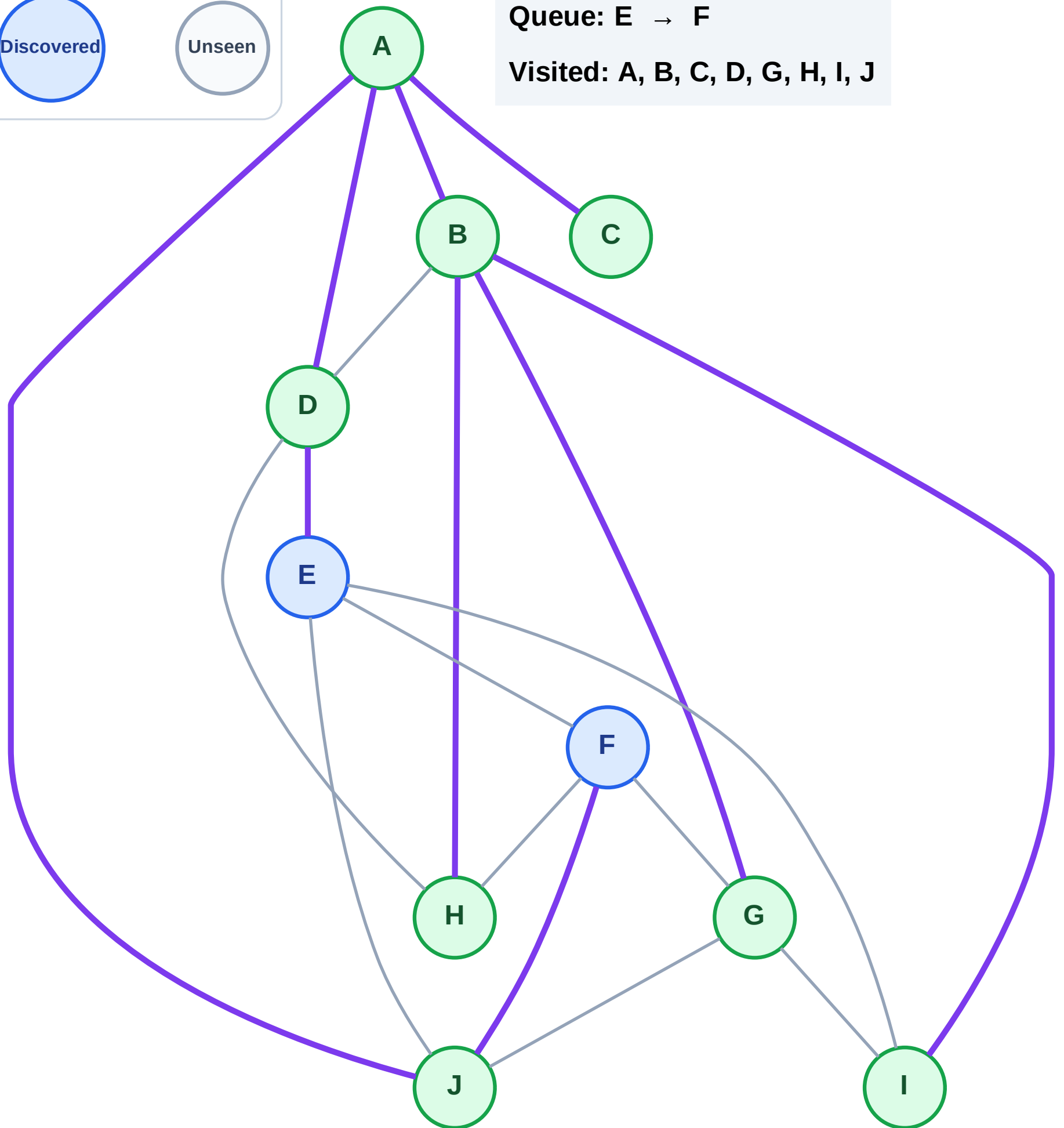
Step 17: No new neighbors from I

Legend



Queue: E → F

Visited: A, B, C, D, G, H, I, J



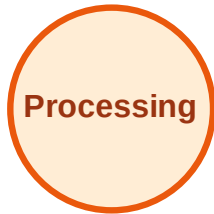
BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



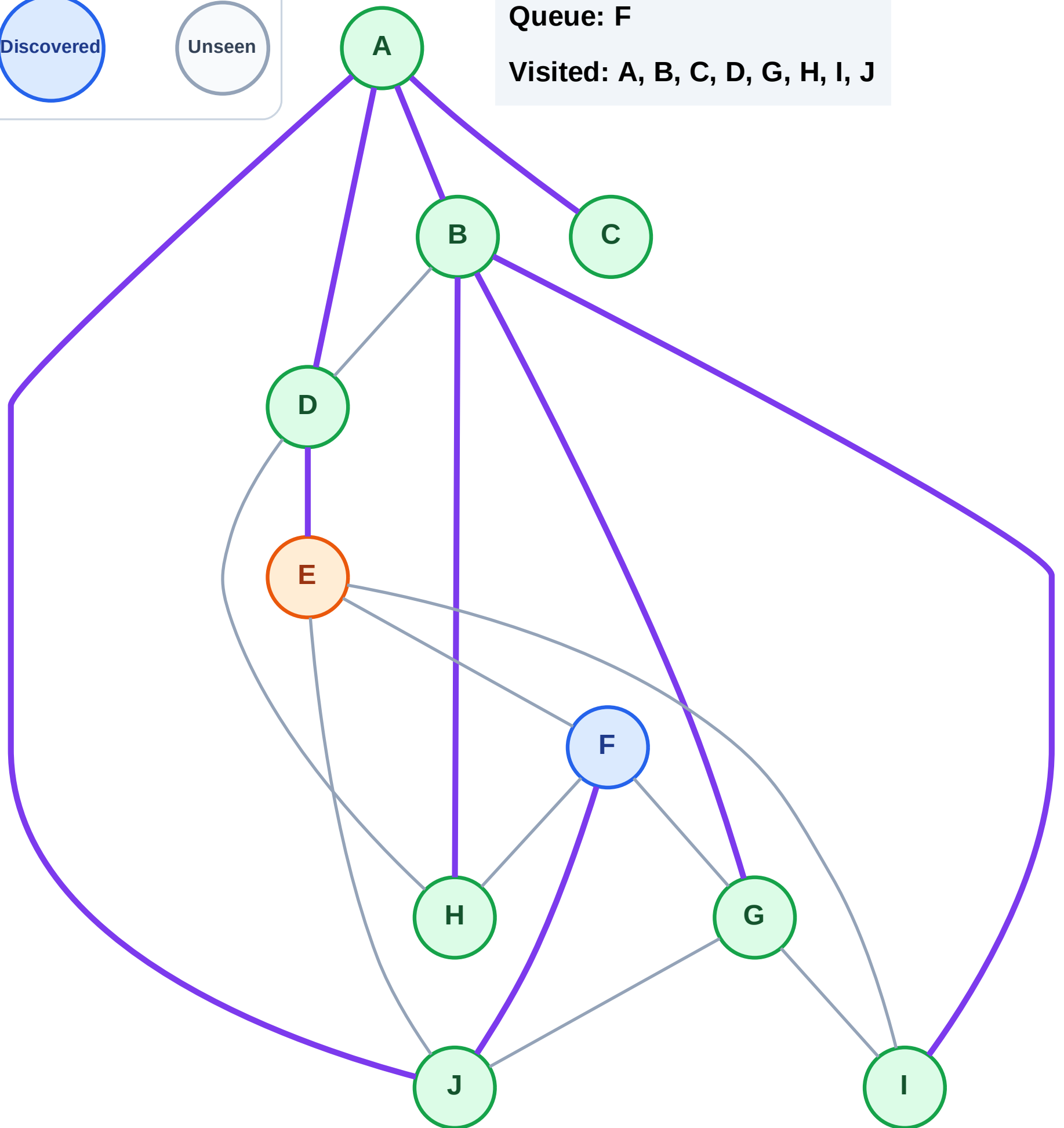
Discovered



Unseen

Queue: F

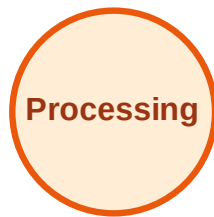
Visited: A, B, C, D, G, H, I, J



BFS Traversal

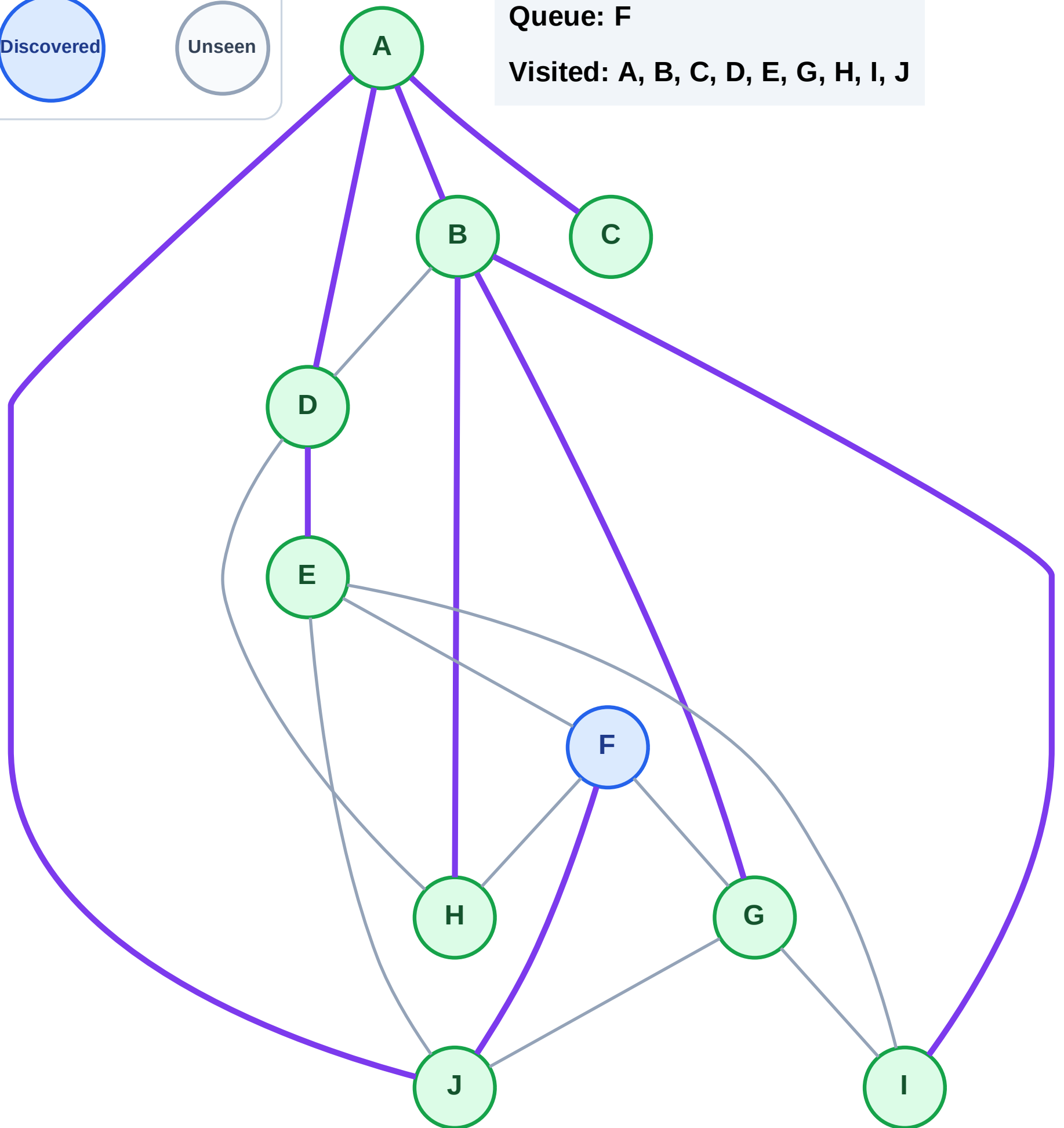
Step 19: No new neighbors from E

Legend



Queue: F

Visited: A, B, C, D, E, G, H, I, J



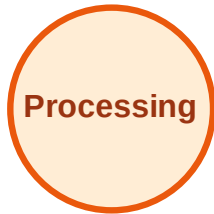
BFS Traversal

Step 20: Process node F

Legend



Complete



Processing



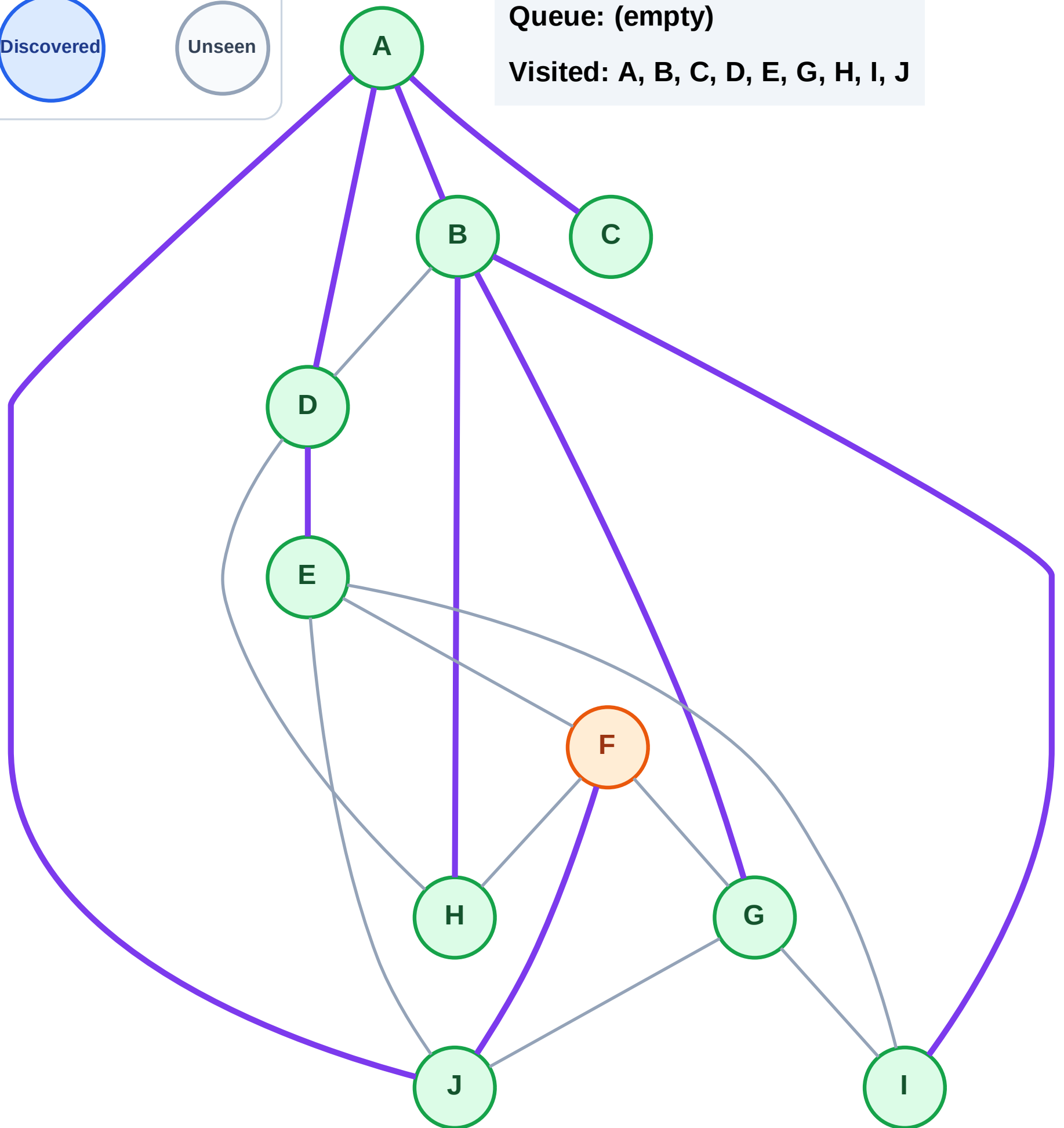
Discovered



Unseen

Queue: (empty)

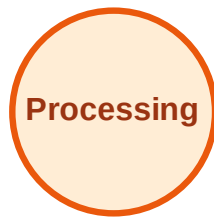
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

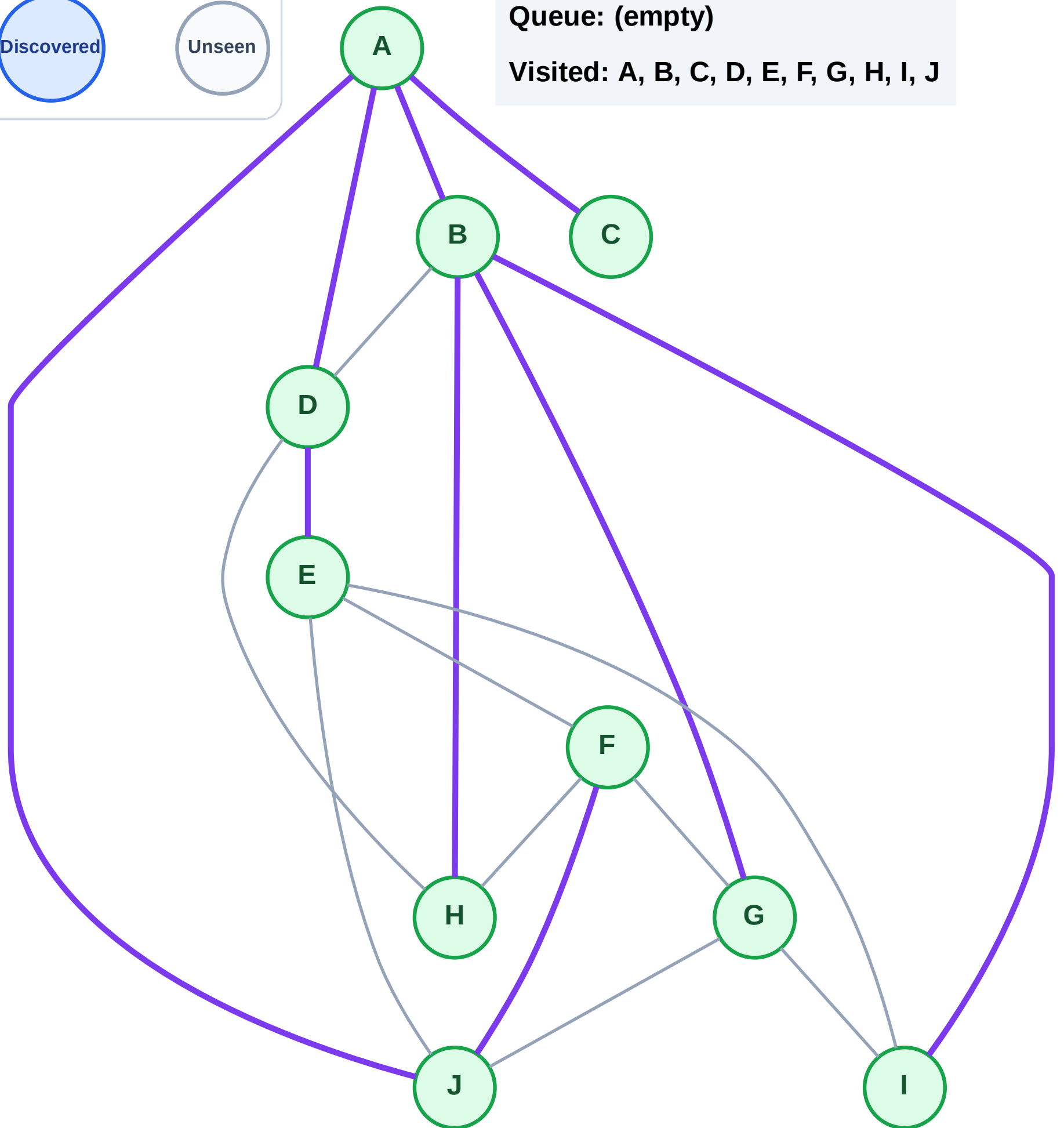
Step 21: No new neighbors from F

Legend



Queue: (empty)

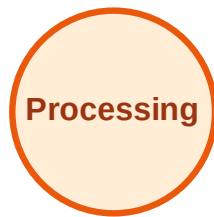
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

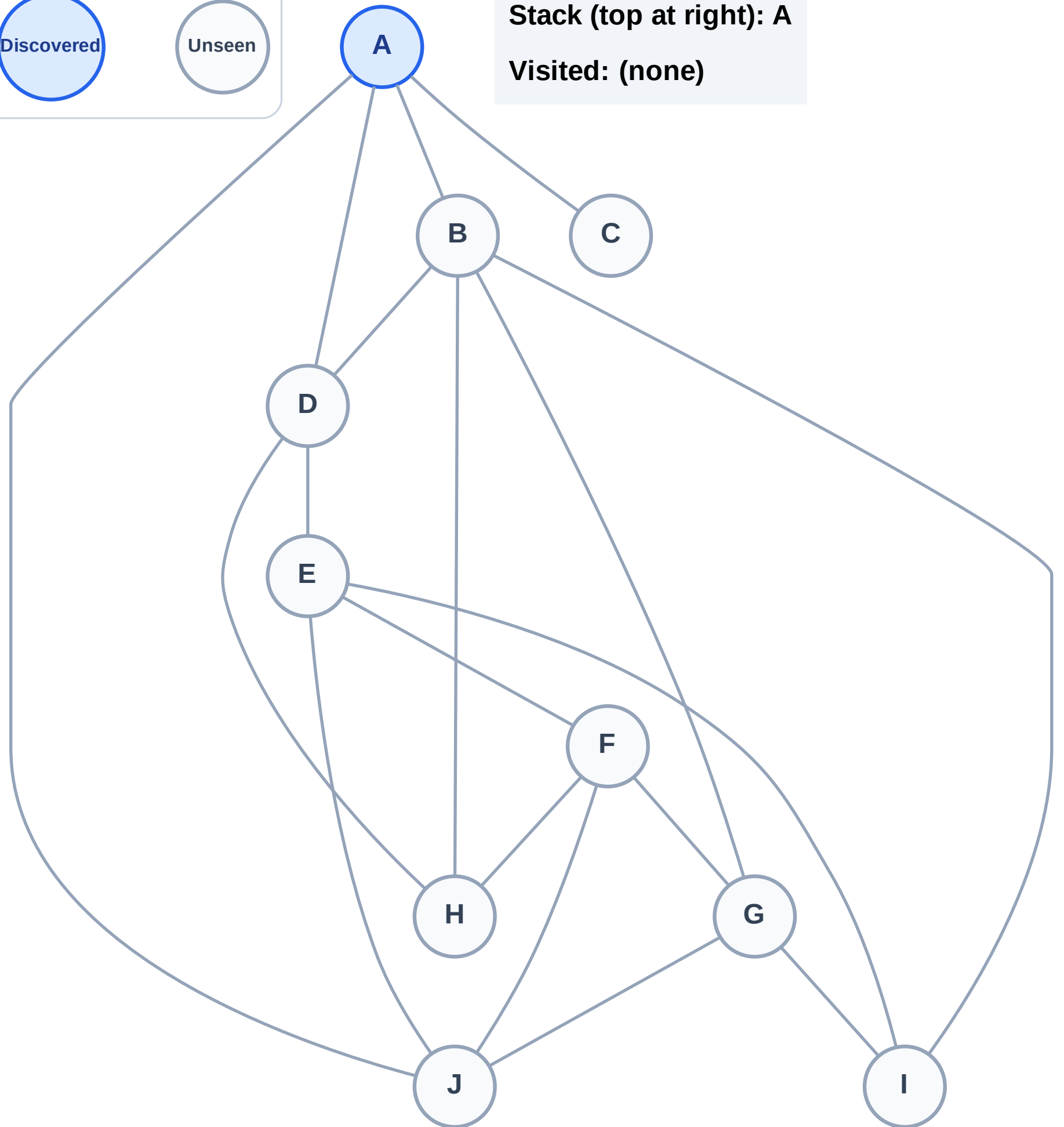
Step 1: Start at A

Legend



Stack (top at right): A

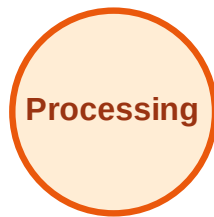
Visited: (none)



DFS Traversal

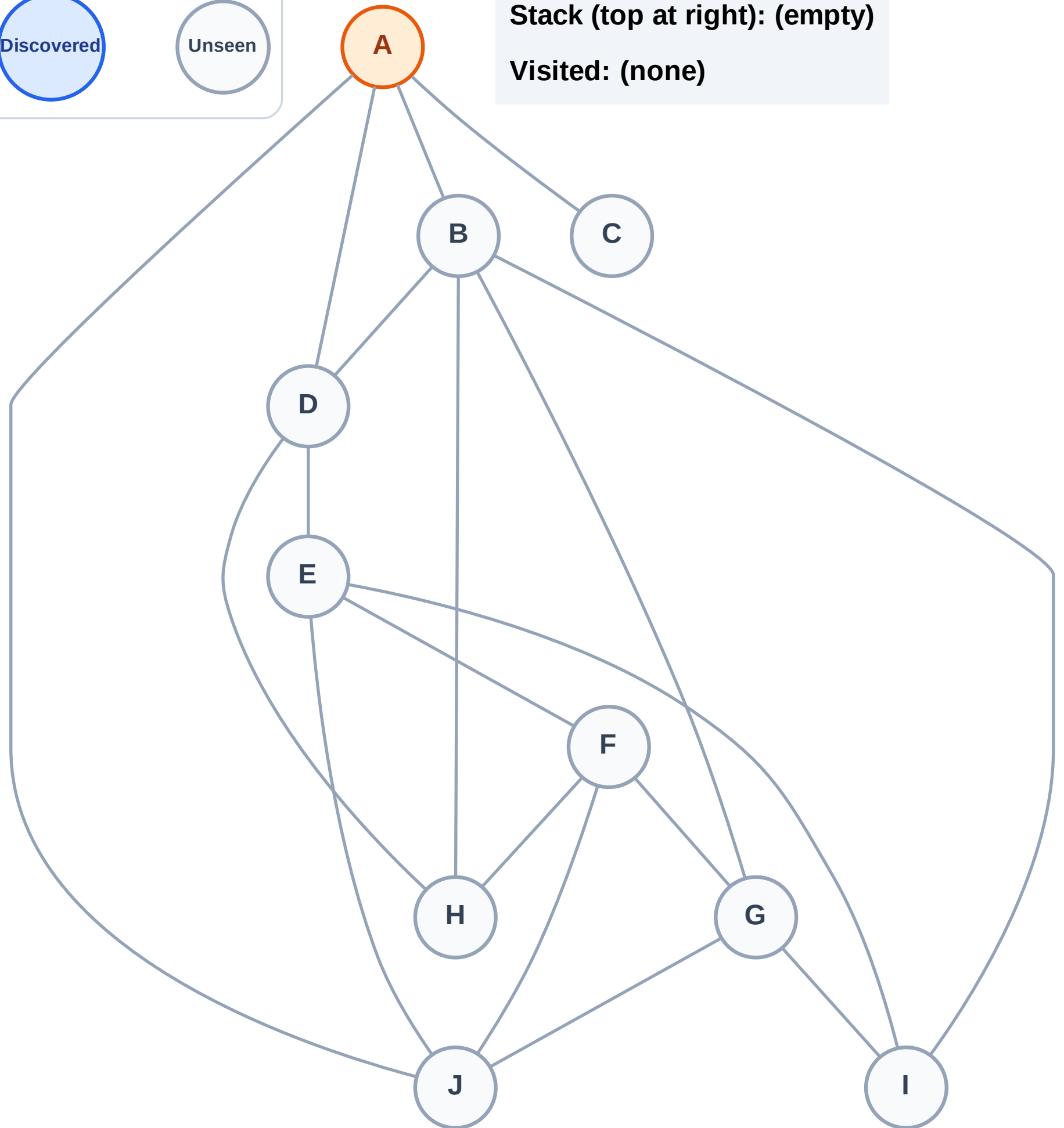
Step 2: Process node A

Legend



Stack (top at right): (empty)

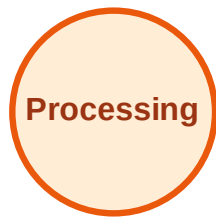
Visited: (none)



DFS Traversal

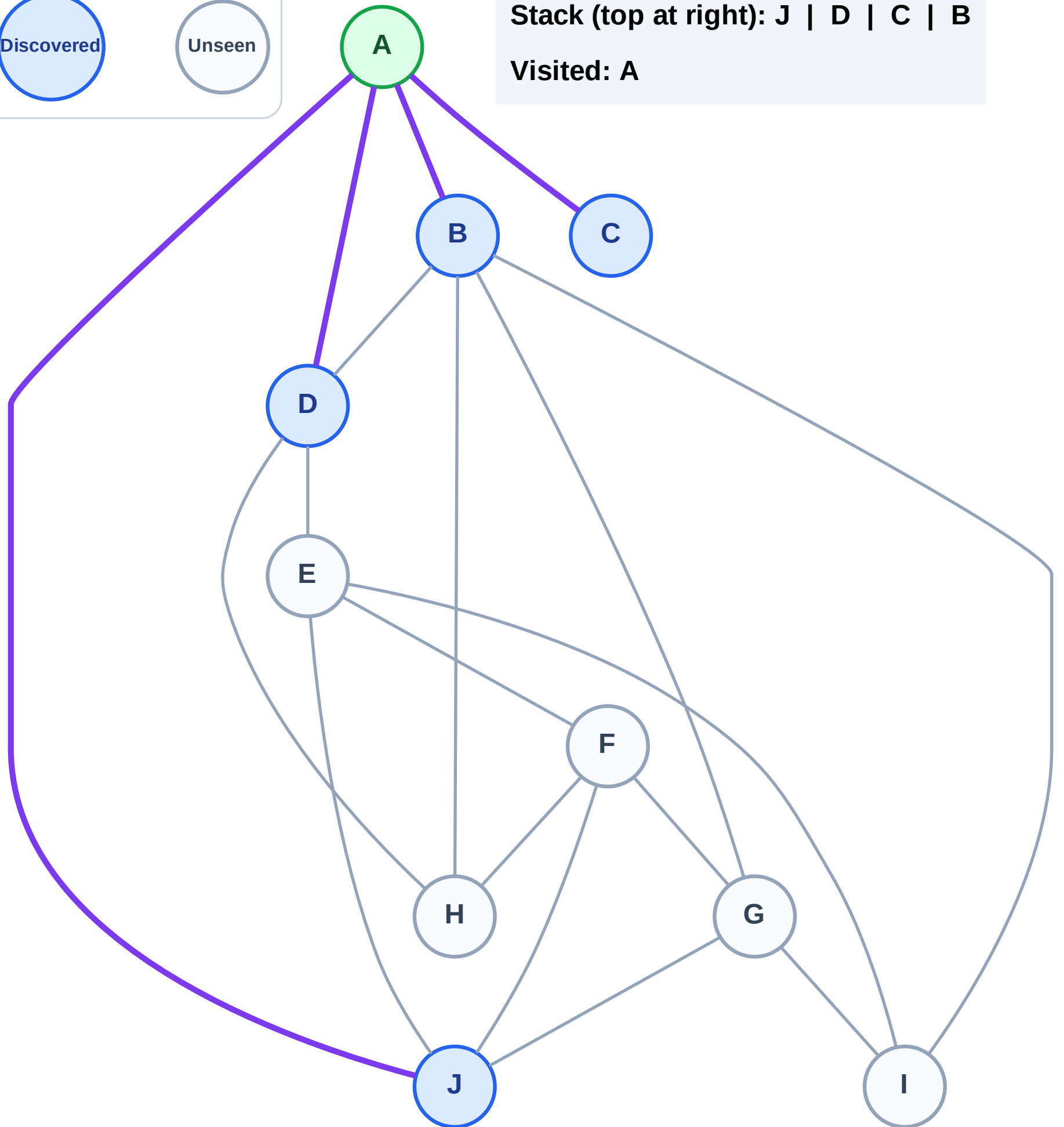
Step 3: Discover J, D, C, B from A

Legend



Stack (top at right): J | D | C | B

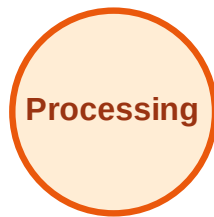
Visited: A



DFS Traversal

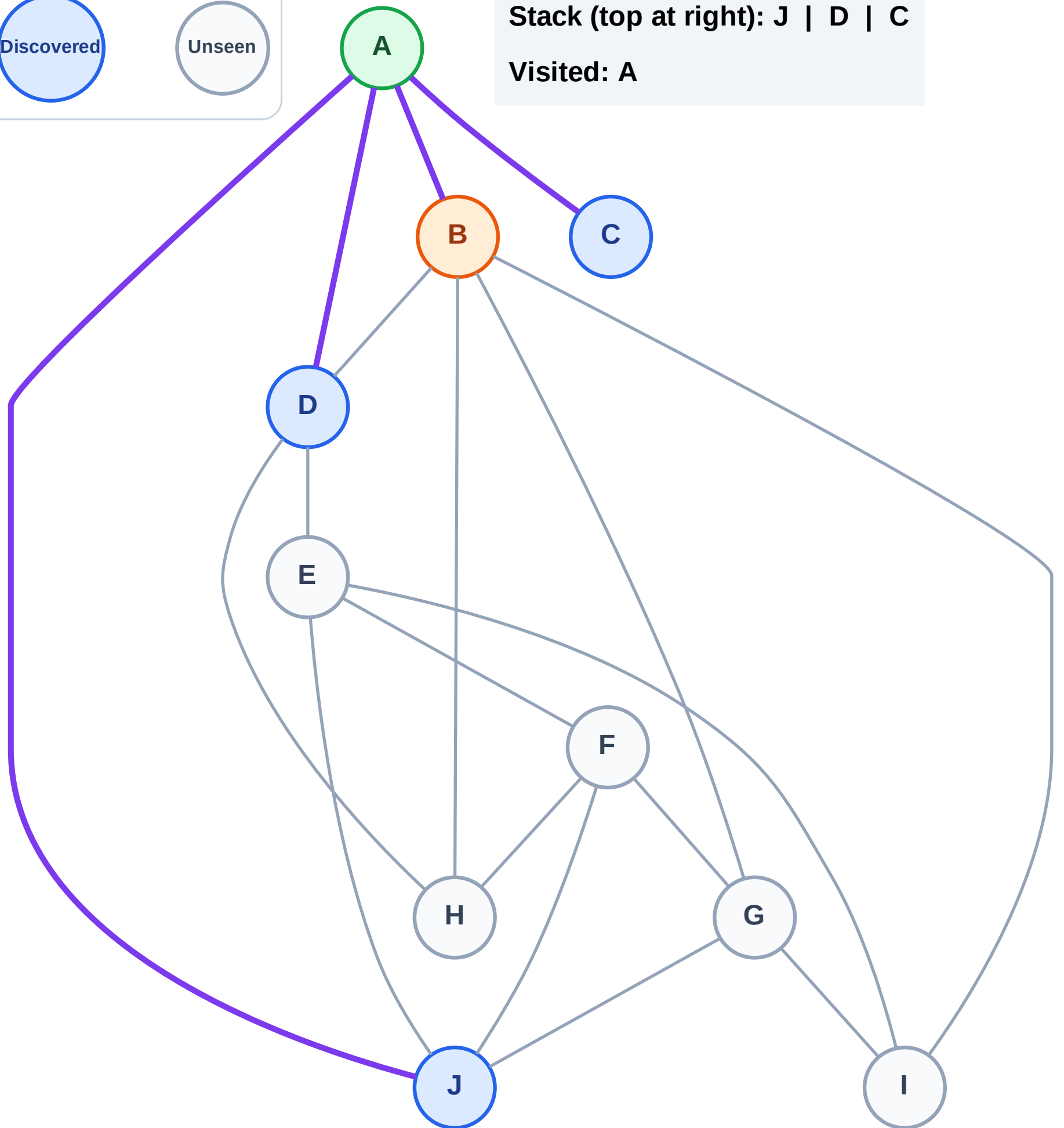
Step 4: Process node B

Legend



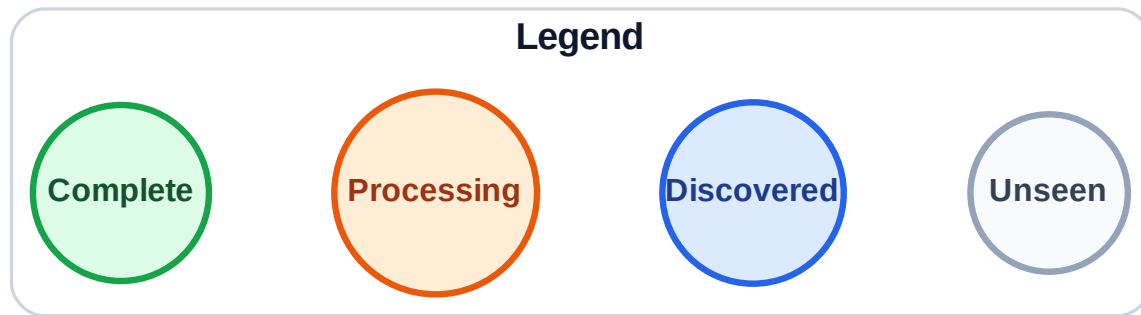
Stack (top at right): J | D | C

Visited: A



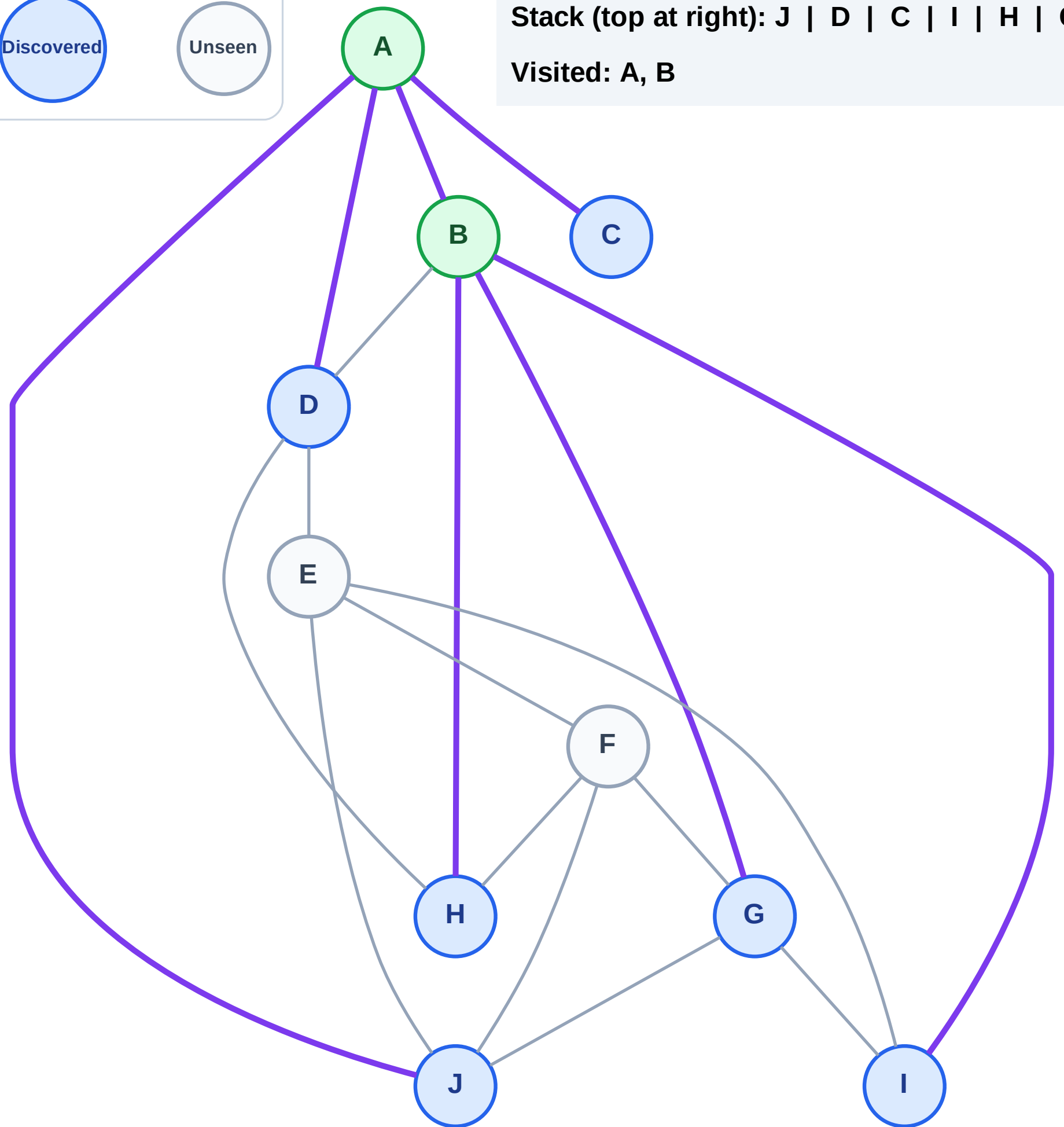
Step 5: Discover I, H, G from B

Step 5: Discover I, H, G from B



Stack (top at right): J | D | C | I | H | G

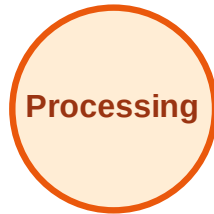
Visited: A, B



DFS Traversal

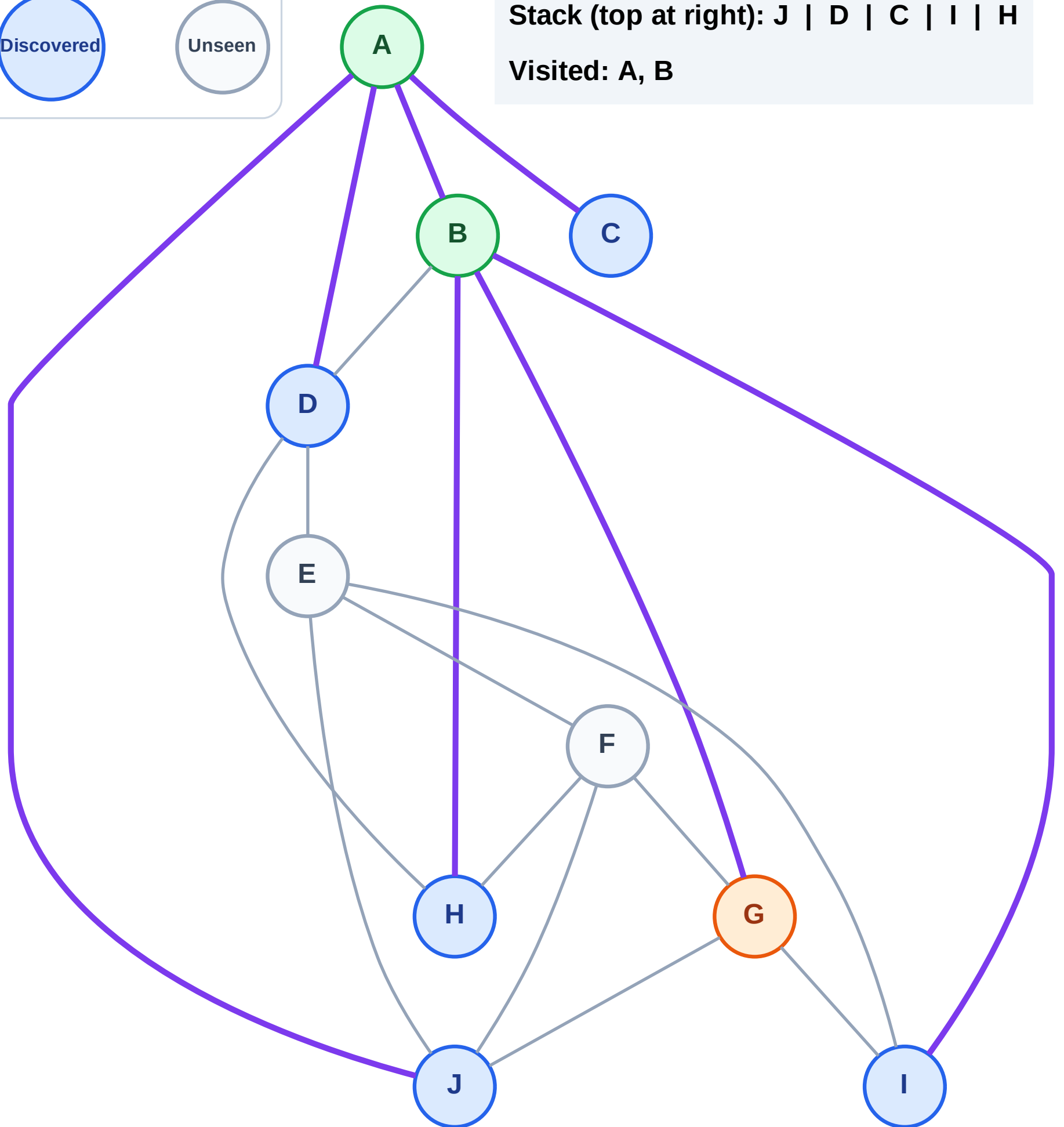
Step 6: Process node G

Legend



Stack (top at right): J | D | C | I | H

Visited: A, B



DFS Traversal

Step 7: Discover F from G

Legend

Complete

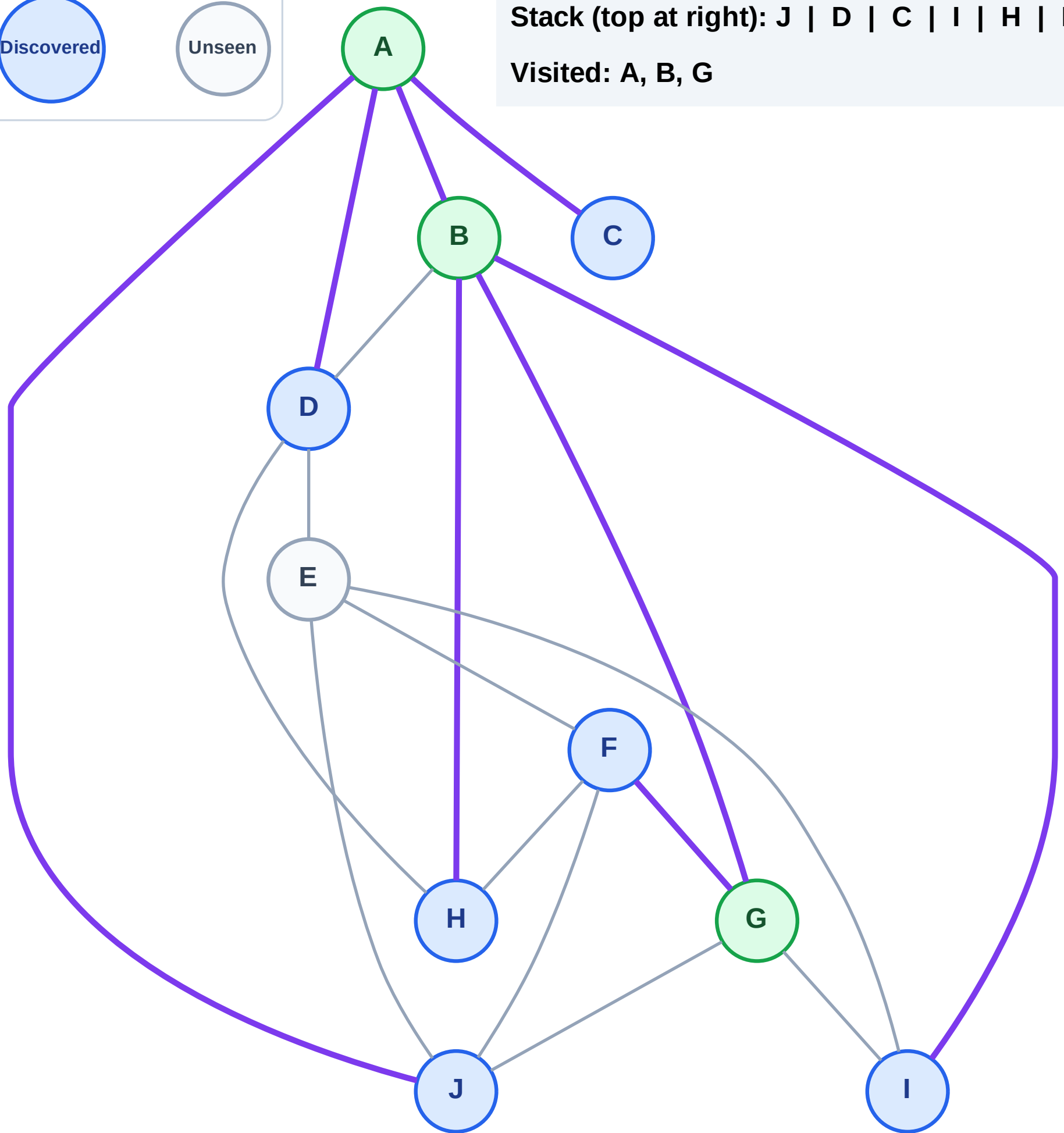
Processing

Discovered

Unseen

Stack (top at right): J | D | C | I | H | F

Visited: A, B, G



DFS Traversal

Step 8: Process node F

Legend

Complete

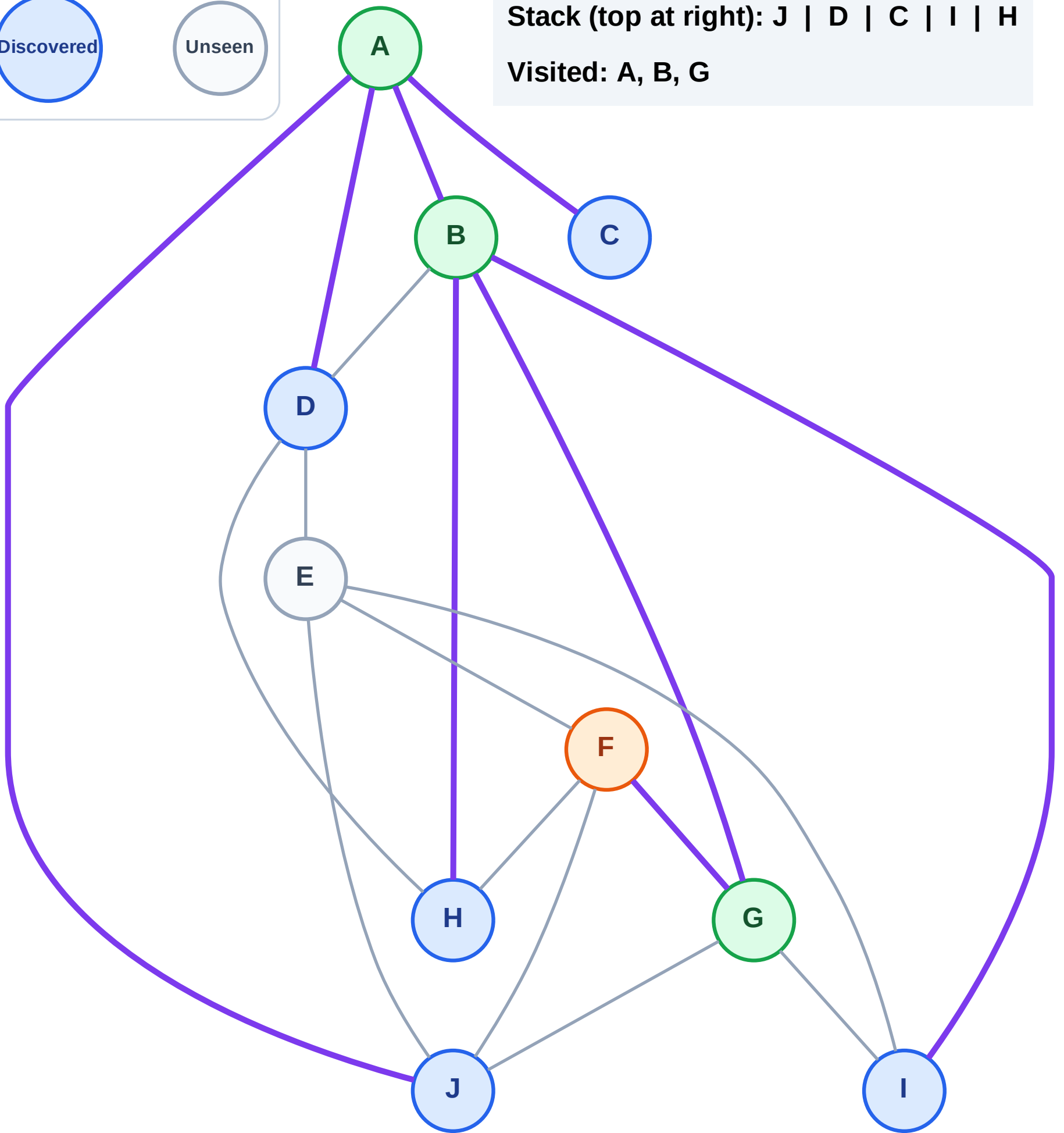
Processing

Discovered

Unseen

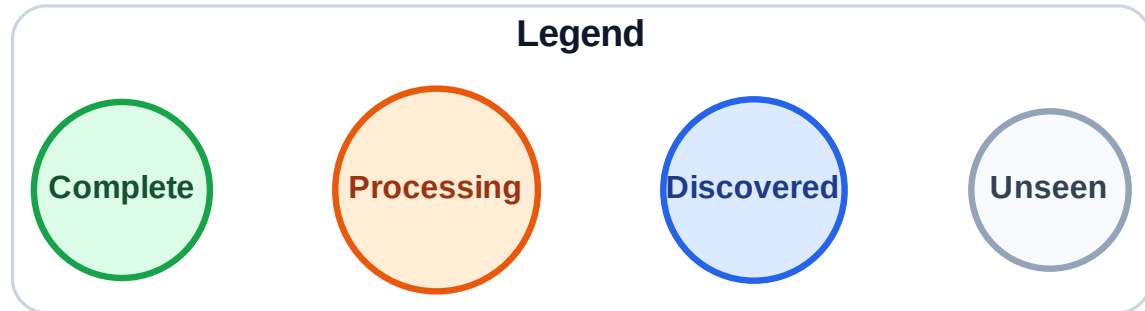
Stack (top at right): J | D | C | I | H

Visited: A, B, G



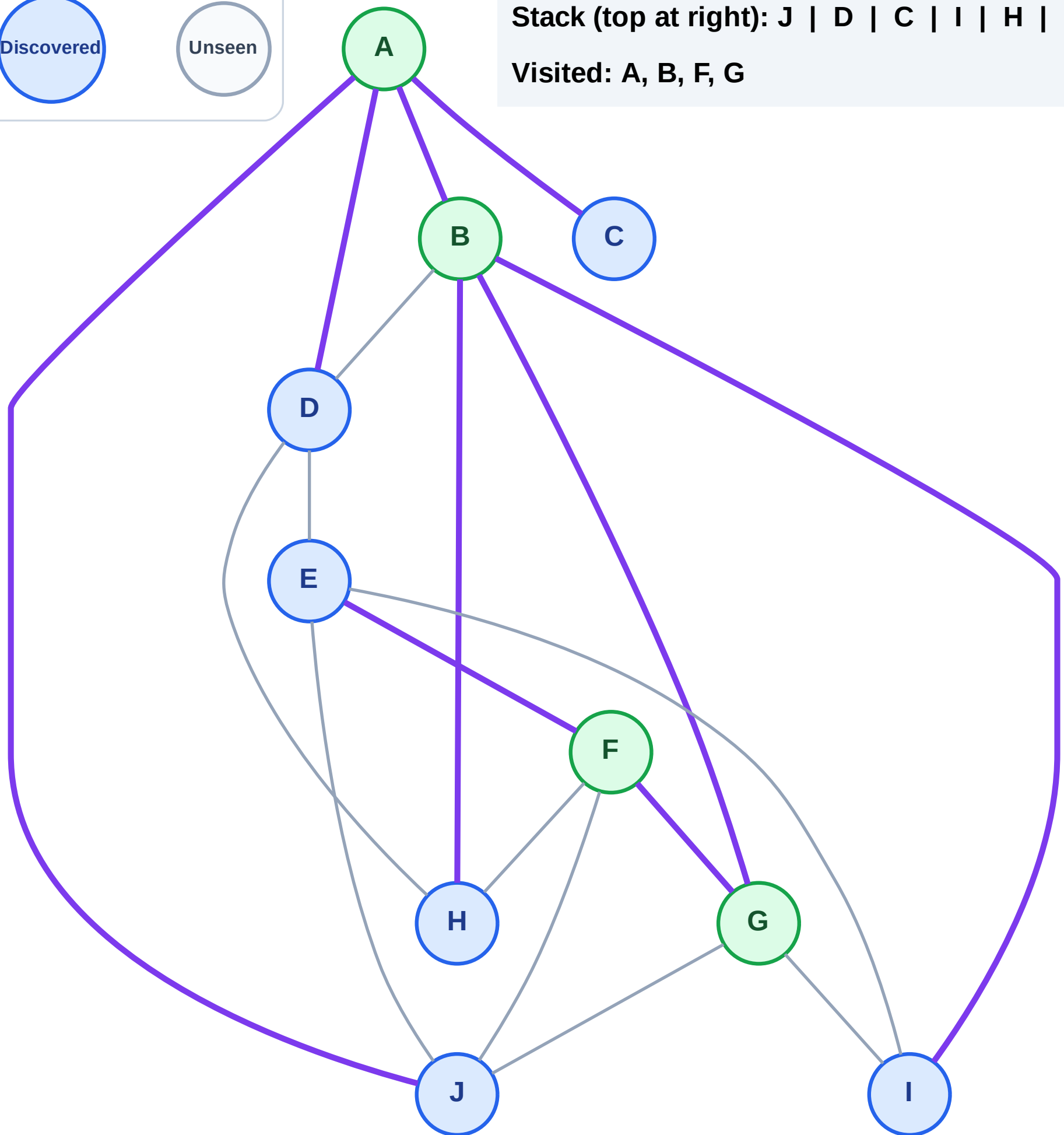
DFS Traversal

Step 9: Discover E from F



Stack (top at right): J | D | C | I | H | E

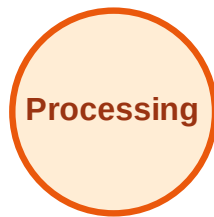
Visited: A, B, F, G



DFS Traversal

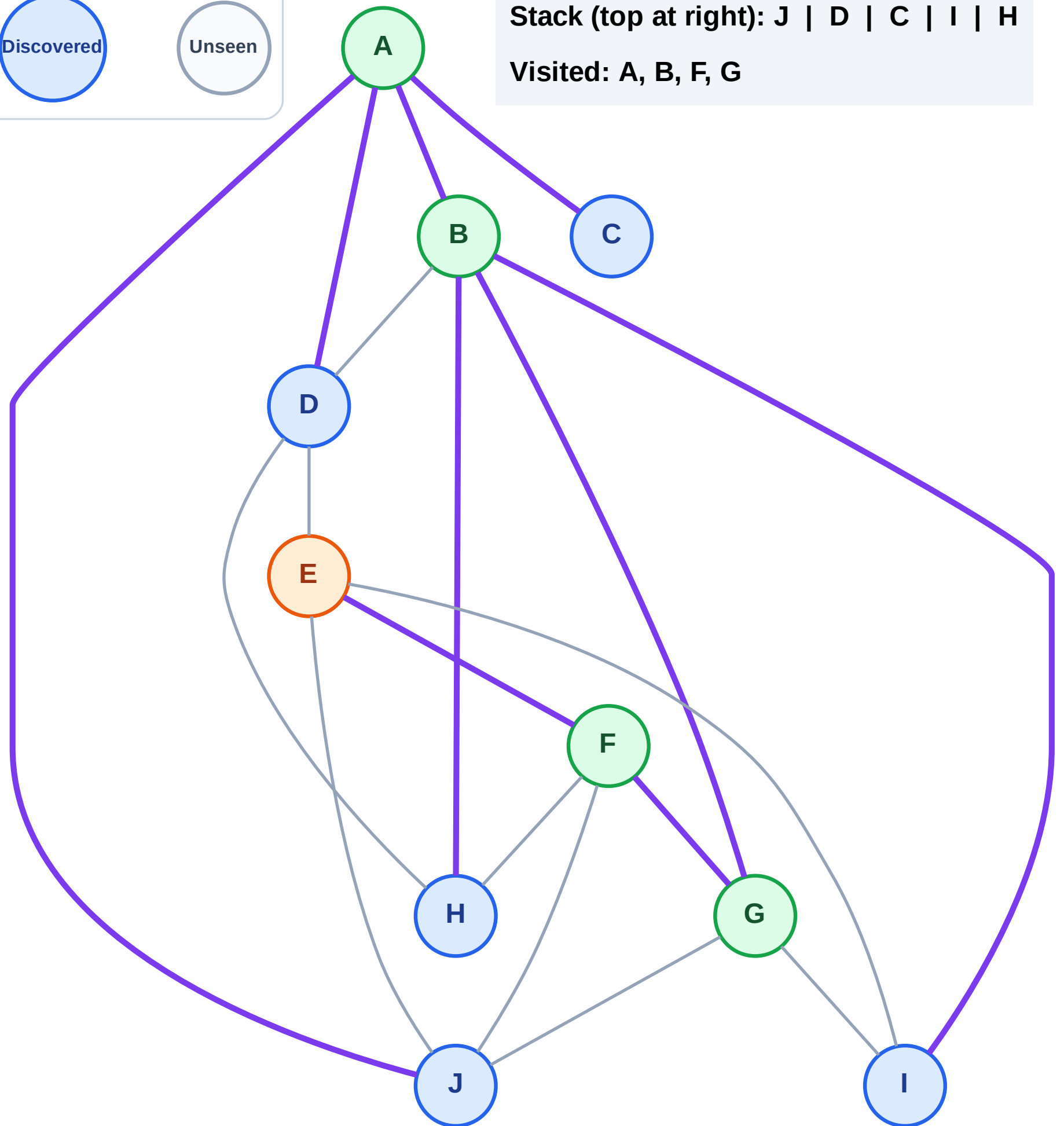
Step 10: Process node E

Legend



Stack (top at right): J | D | C | I | H

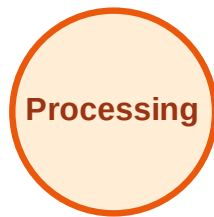
Visited: A, B, F, G



DFS Traversal

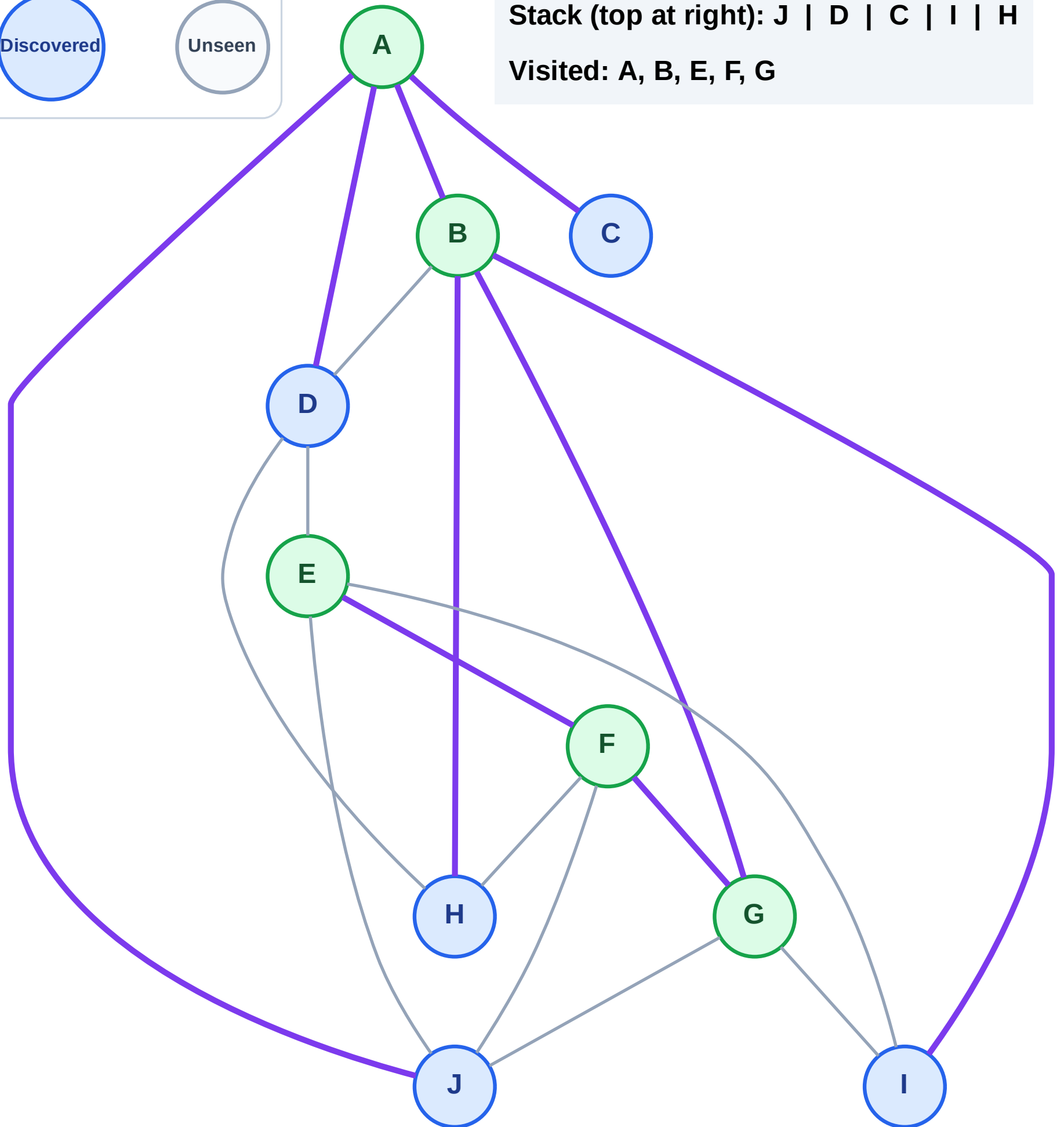
Step 11: No new neighbors from E

Legend



Stack (top at right): J | D | C | I | H

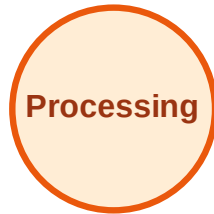
Visited: A, B, E, F, G



DFS Traversal

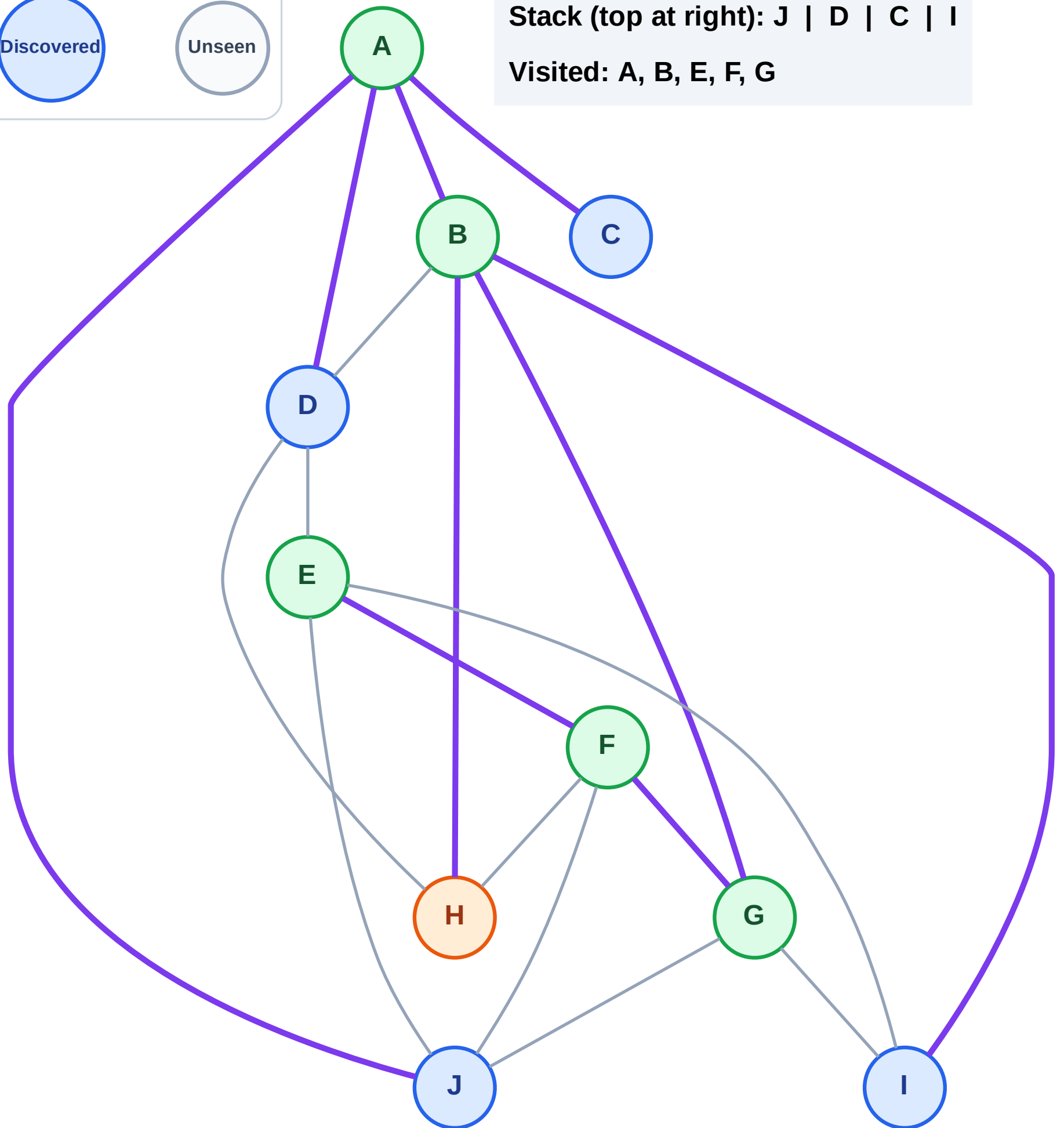
Step 12: Process node H

Legend



Stack (top at right): J | D | C | I

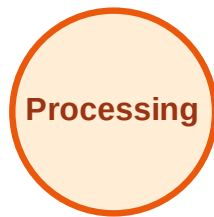
Visited: A, B, E, F, G



DFS Traversal

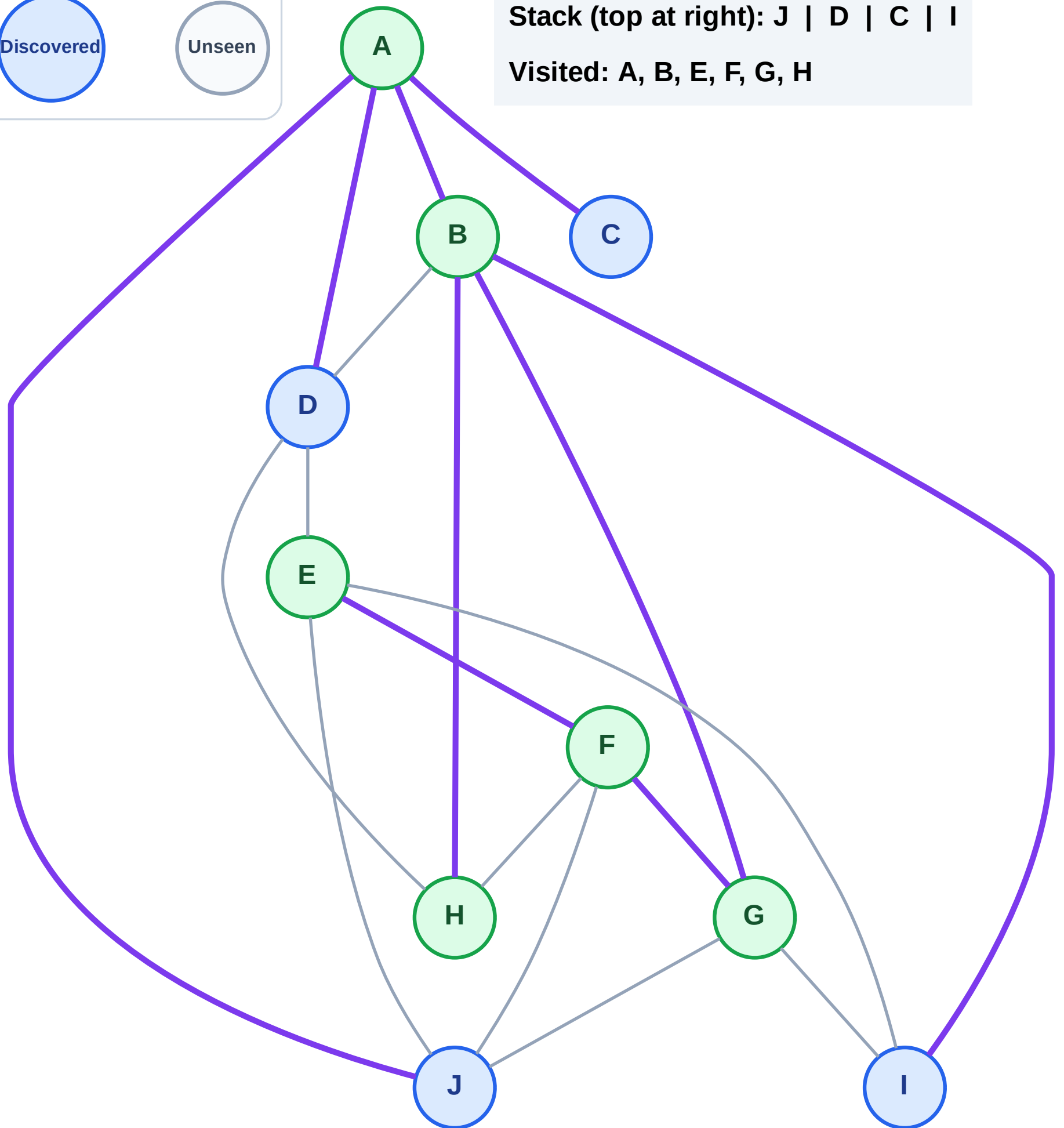
Step 13: No new neighbors from H

Legend



Stack (top at right): J | D | C | I

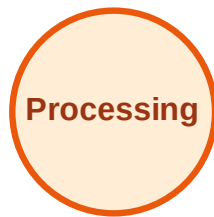
Visited: A, B, E, F, G, H



DFS Traversal

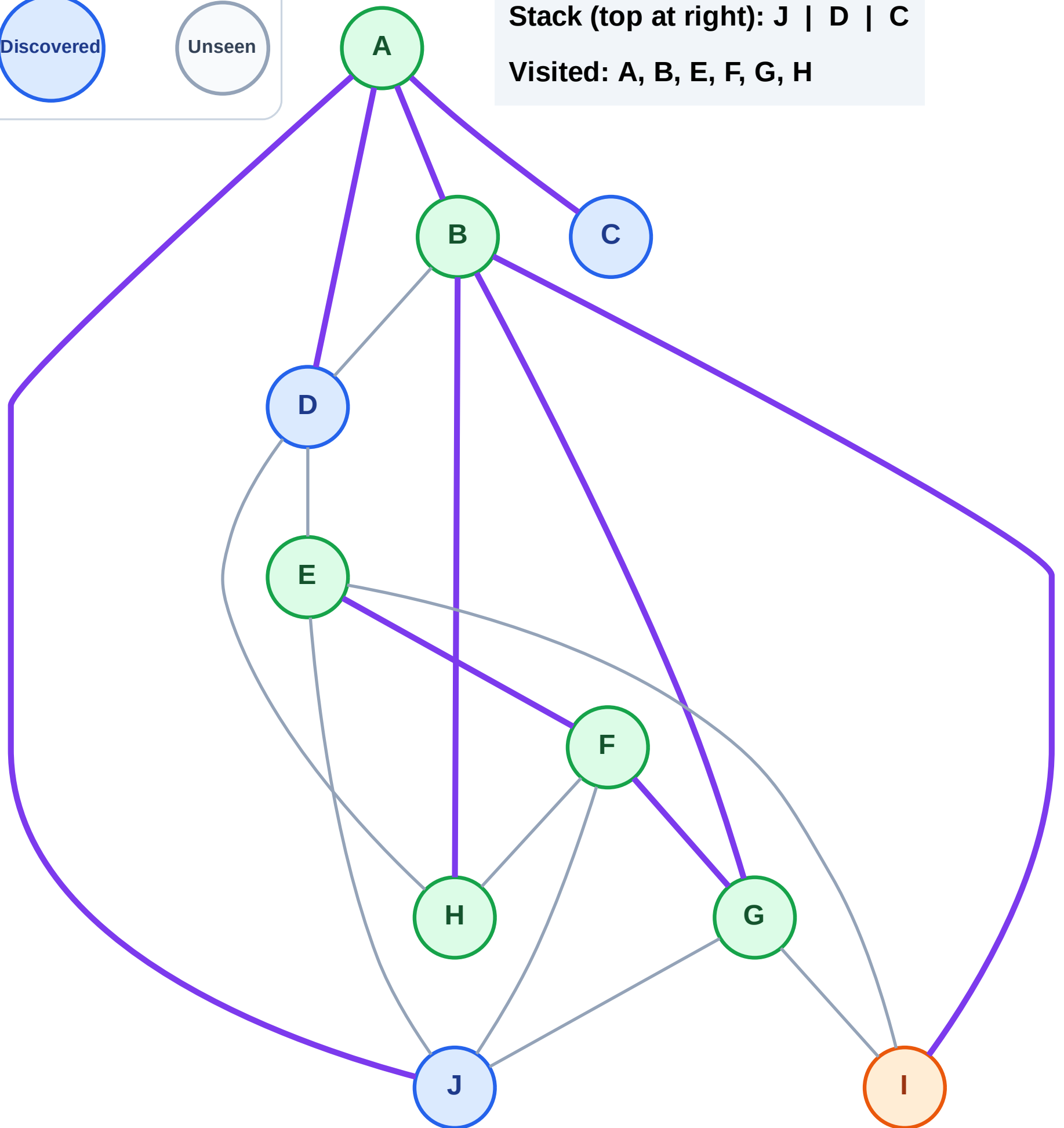
Step 14: Process node I

Legend



Stack (top at right): J | D | C

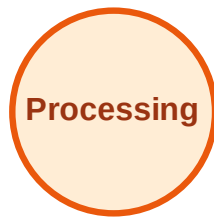
Visited: A, B, E, F, G, H



DFS Traversal

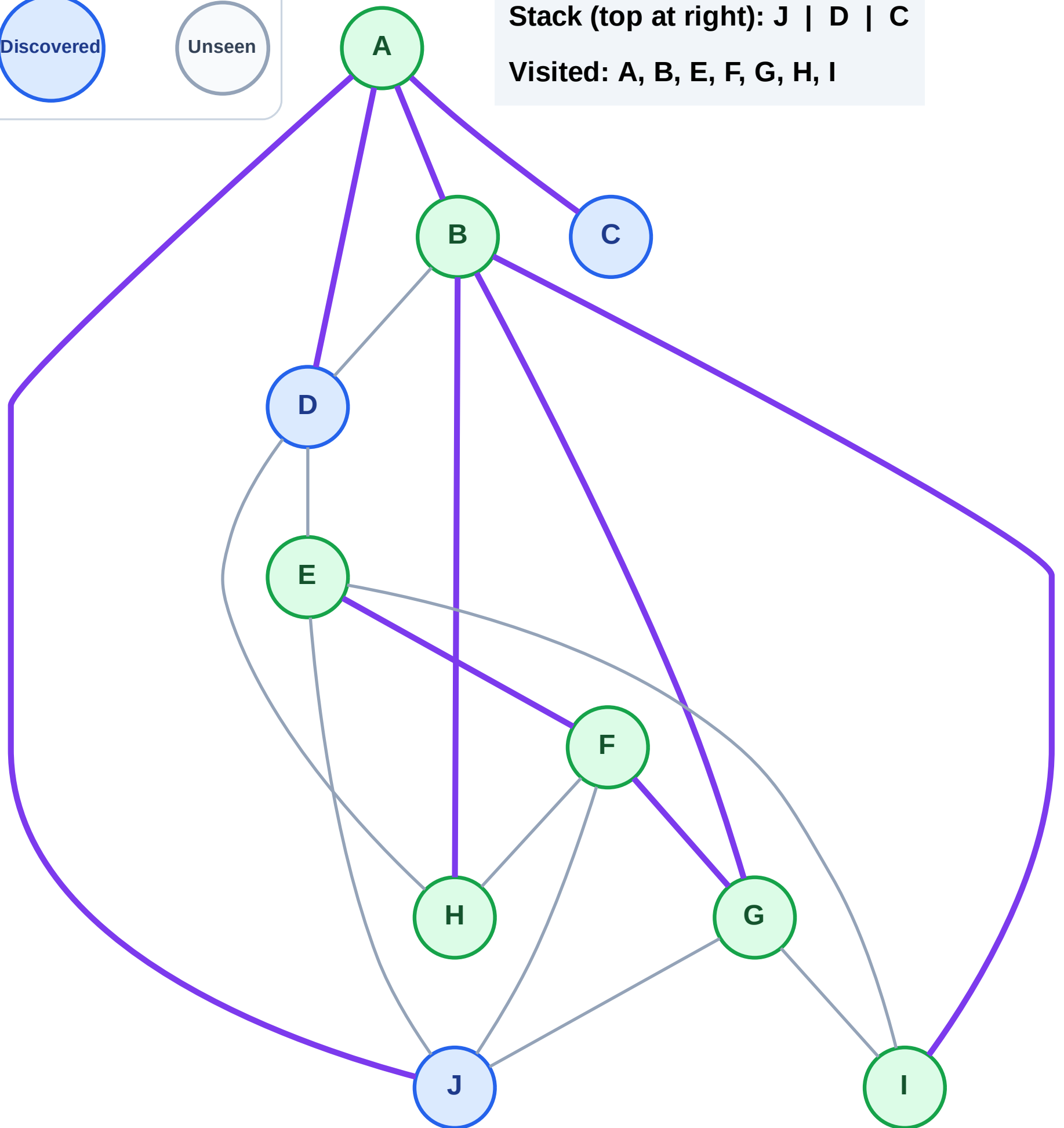
Step 15: No new neighbors from I

Legend



Stack (top at right): J | D | C

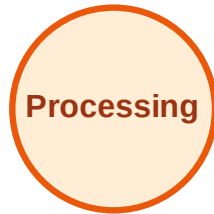
Visited: A, B, E, F, G, H, I



DFS Traversal

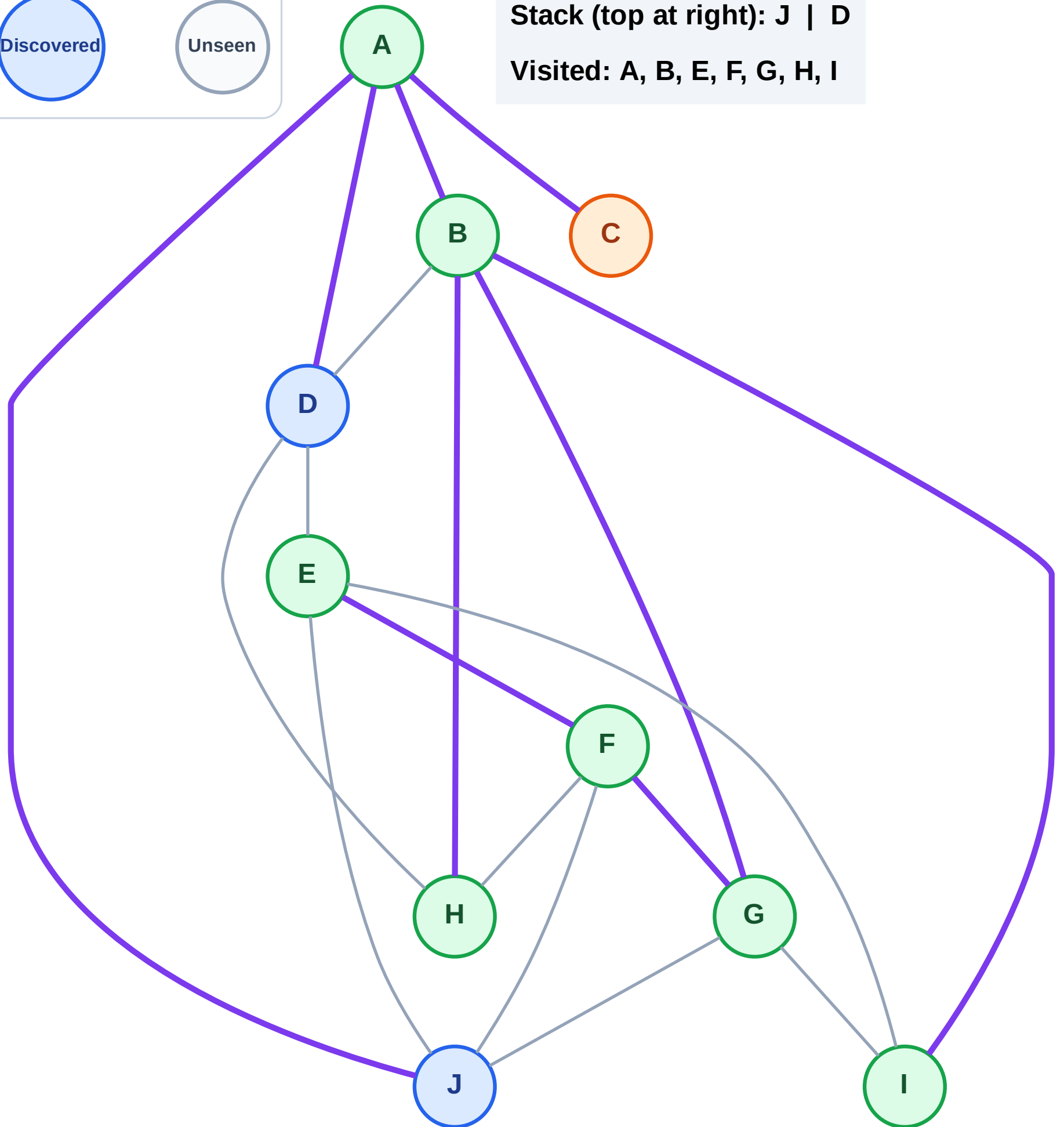
Step 16: Process node C

Legend



Stack (top at right): J | D

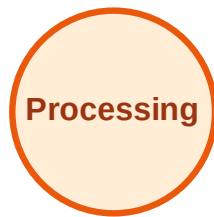
Visited: A, B, E, F, G, H, I



DFS Traversal

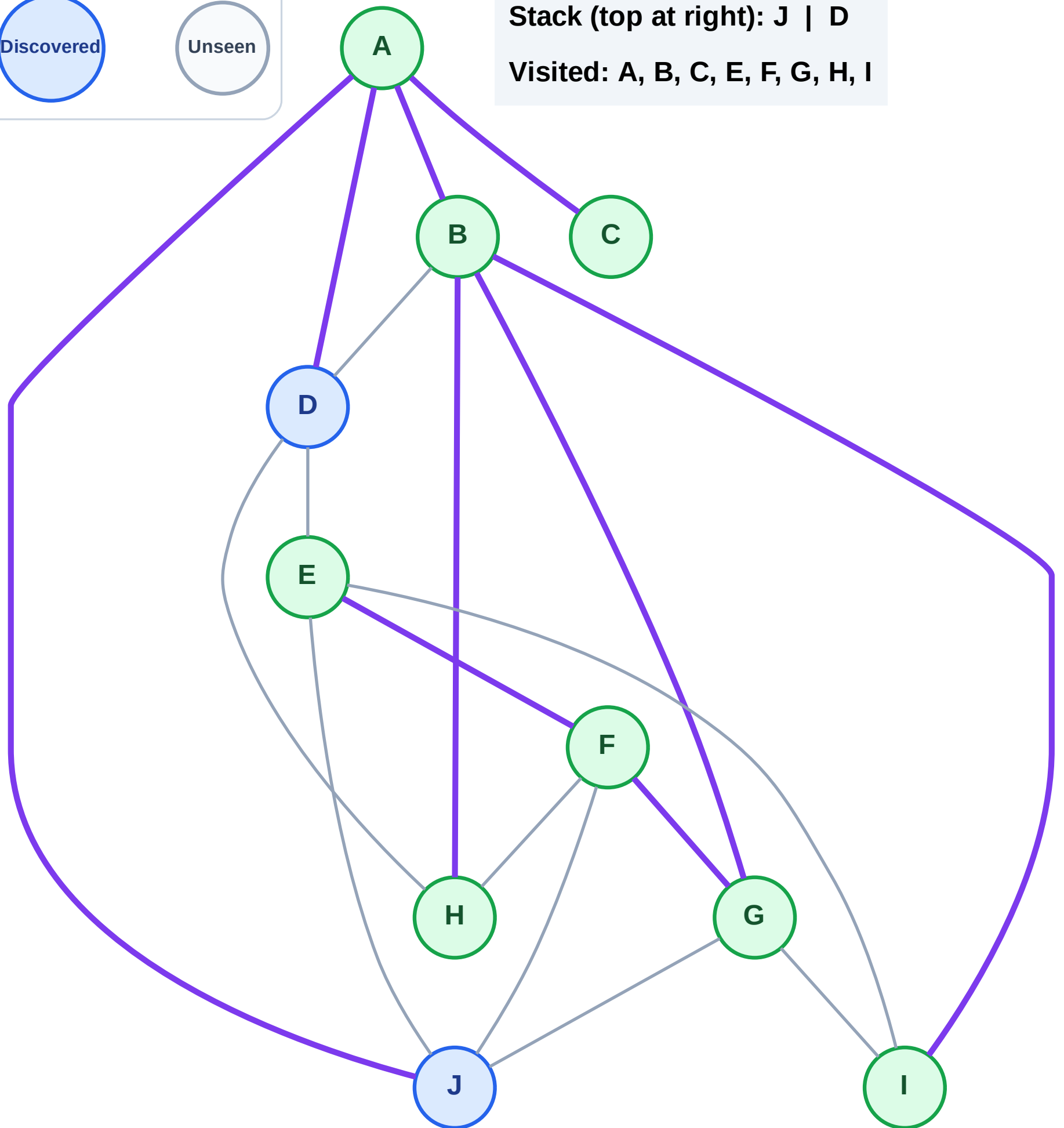
Step 17: No new neighbors from C

Legend



Stack (top at right): J | D

Visited: A, B, C, E, F, G, H, I



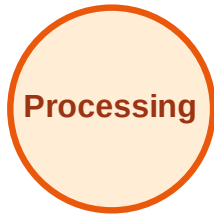
DFS Traversal

Step 18: Process node D

Legend



Complete



Processing



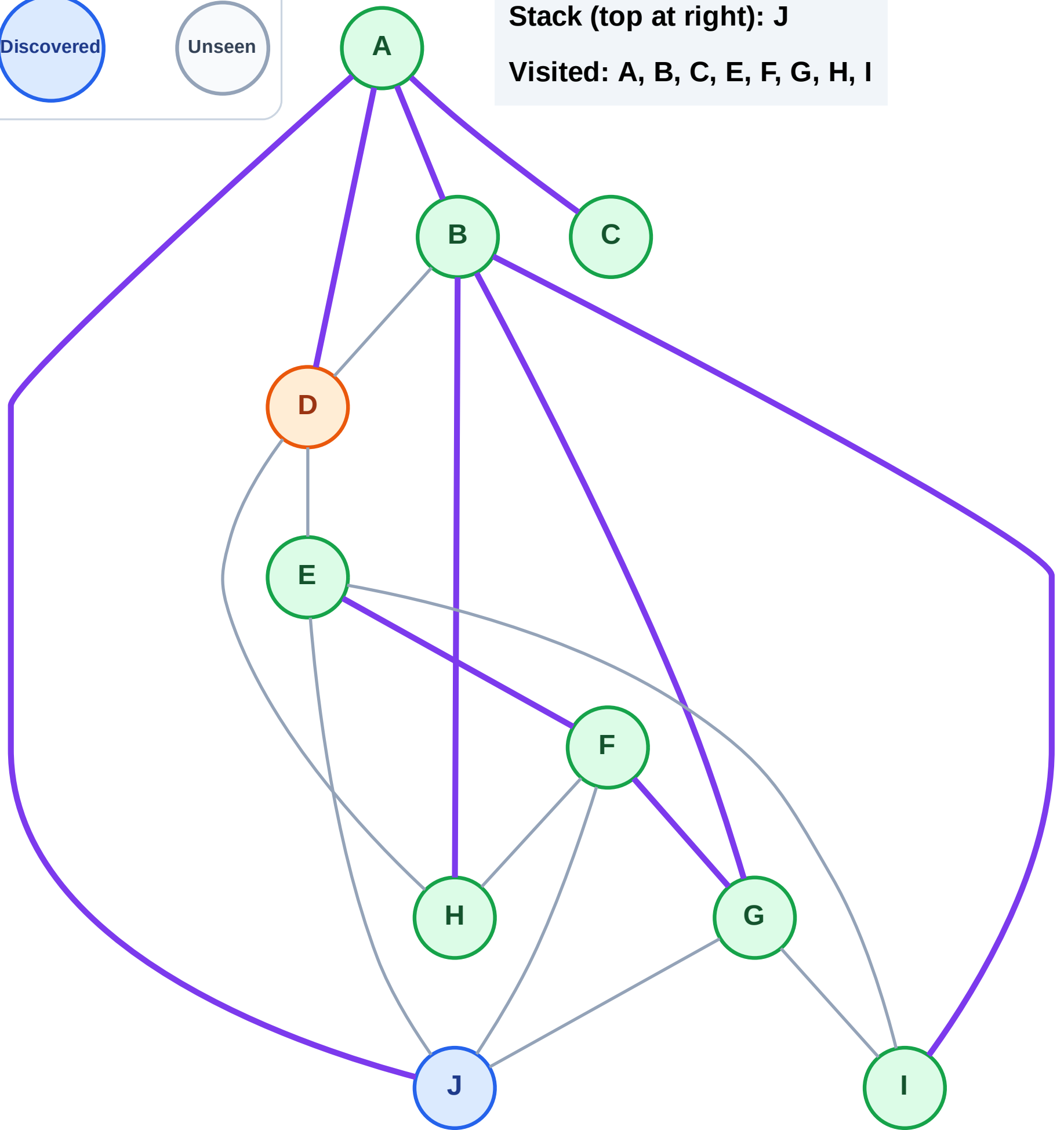
Discovered



Unseen

Stack (top at right): J

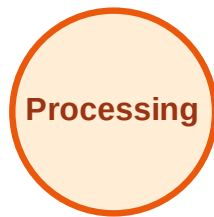
Visited: A, B, C, E, F, G, H, I



DFS Traversal

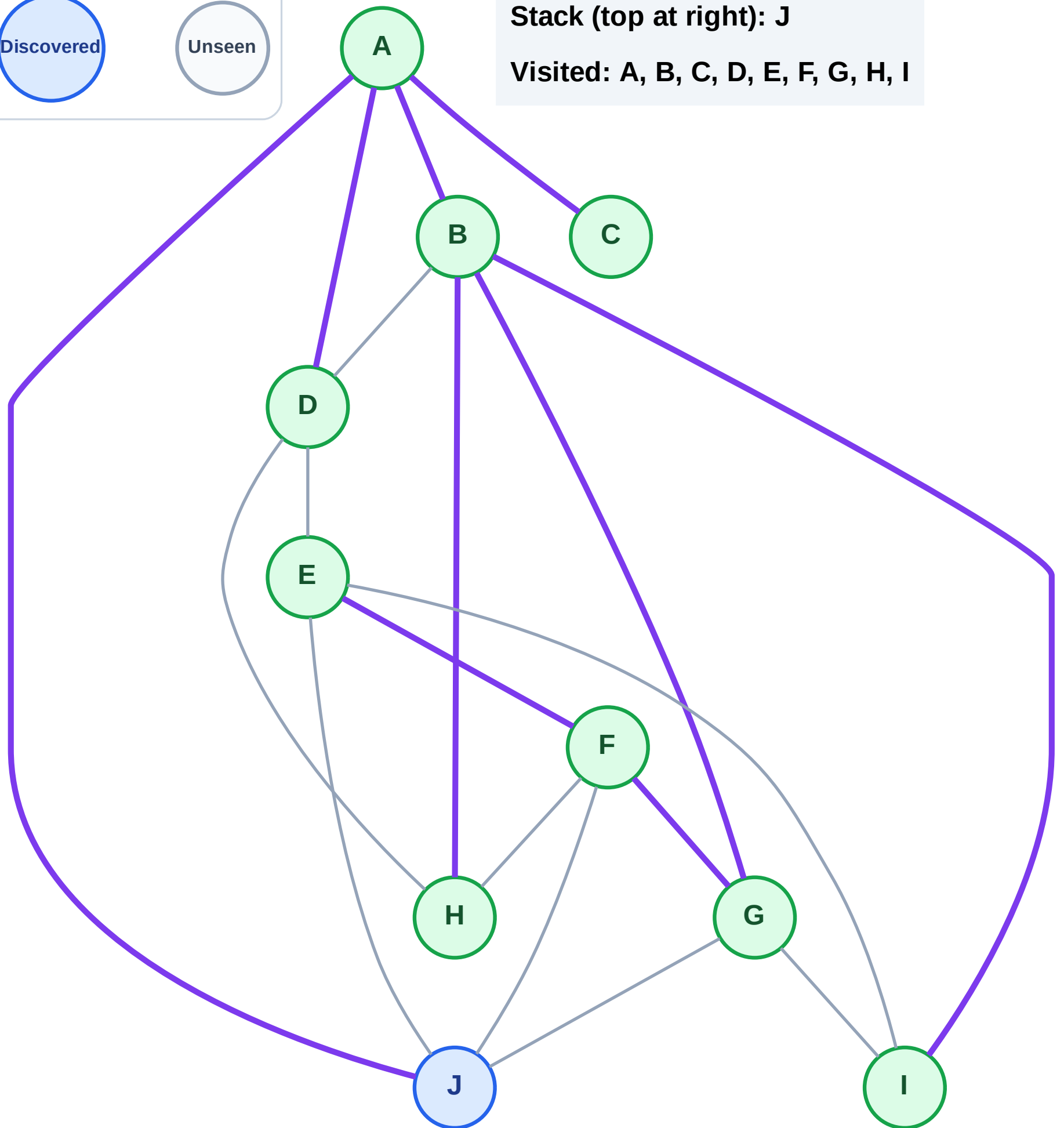
Step 19: No new neighbors from D

Legend



Stack (top at right): J

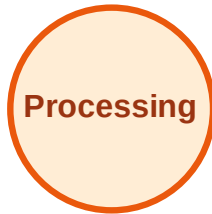
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

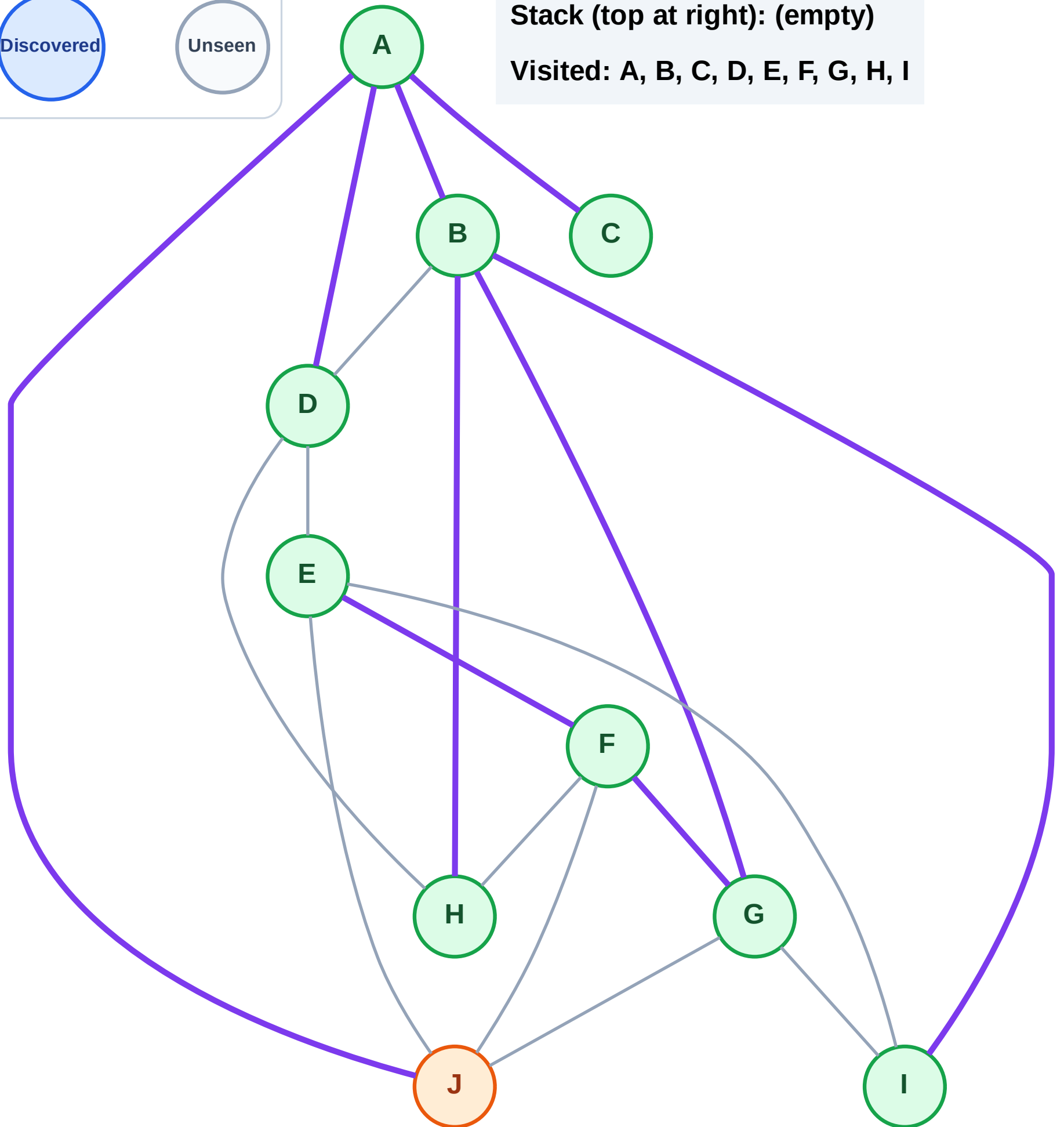
Step 20: Process node J

Legend



Stack (top at right): (empty)

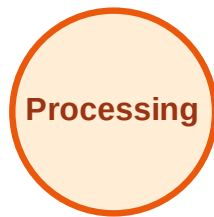
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

